Sensors

to perceive the environment

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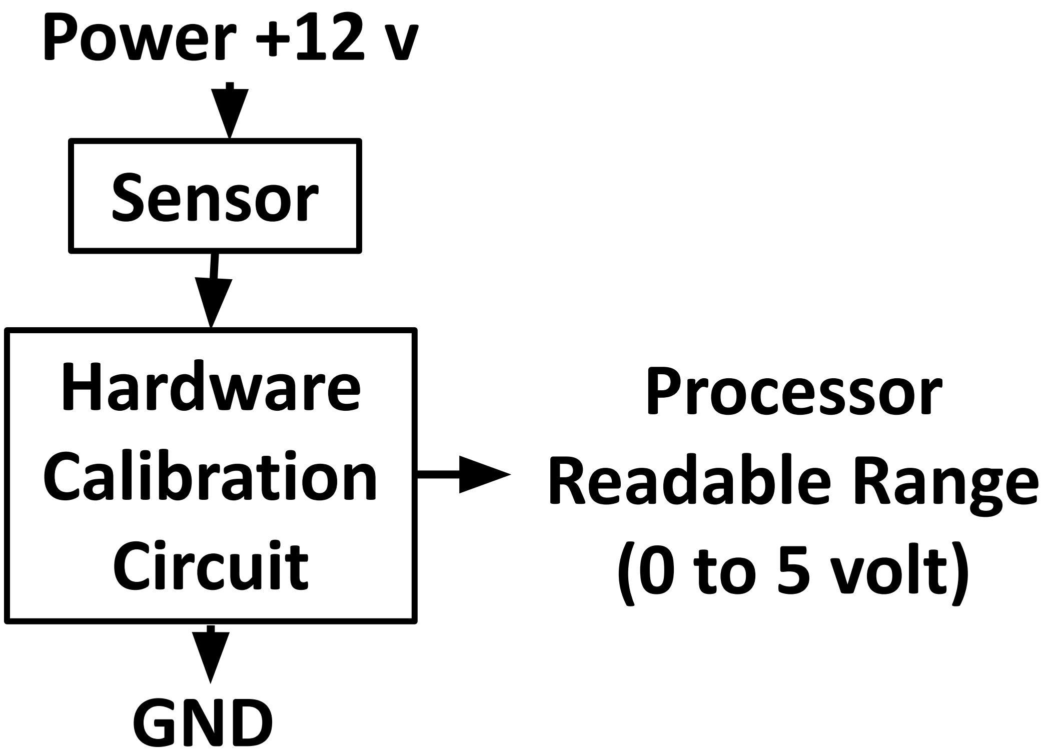
Hardware Calibration

- Processor can read either digital or Analog value
  - Digital should be “0” (Gnd) or “1” (>3.5 Volt)
  - Analog should be in a range according to the analog reference voltage (Usually 0 volt to 5 volt).
    - Analog Voltage □ ADC □ Numerical Value (Usually 0 to 255 (8 bit Binary Nmber))

- **So, a sensor input must be calibrated into a readable range of Processor with a circuit.**

Hardware Calibration Block Diagram



To get efficient result, hardware Calibration should be in the way so that the lower value goes to lower range and upper value goes to upper range.

Sensors

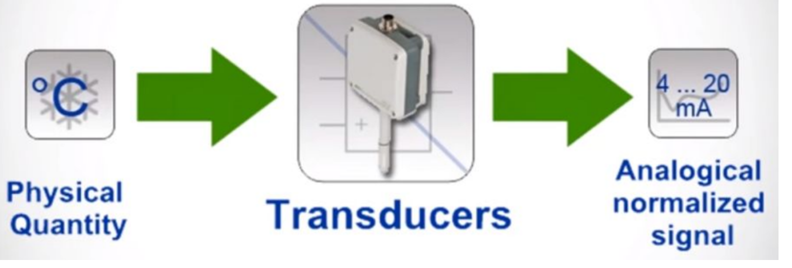
- By Material
  - Thermistor
  - Piezoelectric
  - Pyro electric material (IR)
  - Electromagnetic
  - Semiconductor
  - Radio active material
  - Capacitive
  - Resistive
  - Conductive
- By Engineering
  - Gyroscope
  - Microphone
  - Camera
  - LiDAR
  - Pulse Oximeter
  - Level Sensor
  - Hall-effect
  - Break-Beam
  - Vibrator
  - Accelarometer

Sensors

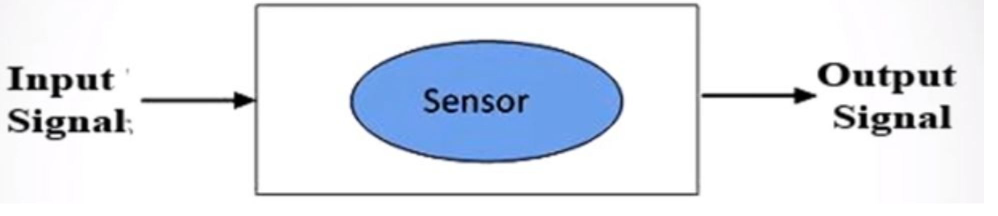
- Active Sensor:
  - Requires emitter
- Passive Sensor:
  - Sense environment directly. No need to emit any signal.

Transducer & Sensor

- Transducer: Convert energy from one form to another
  - Pressure, Piezoelectric, Ultra-sound, Temperature



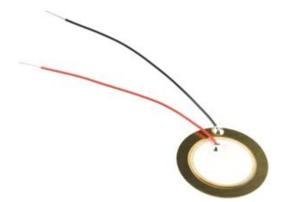
- Sensor: Detect physical quantity and convert into electrical Signal



Transducer

- Pressure (Refrigerator and AC)

- Piezoelectric (Buzzer or lighter)



- Ultra-sound (Ultrasonography, Sonar)

- Temperature (Engines and boilers)

Sensors

- Light based Sensors
- Magnetic & Electro-Magnetic sensor
- Sound based Sensors
- Some Modern Sensors
- Water conductivity based sensors
- Underwater Sensors

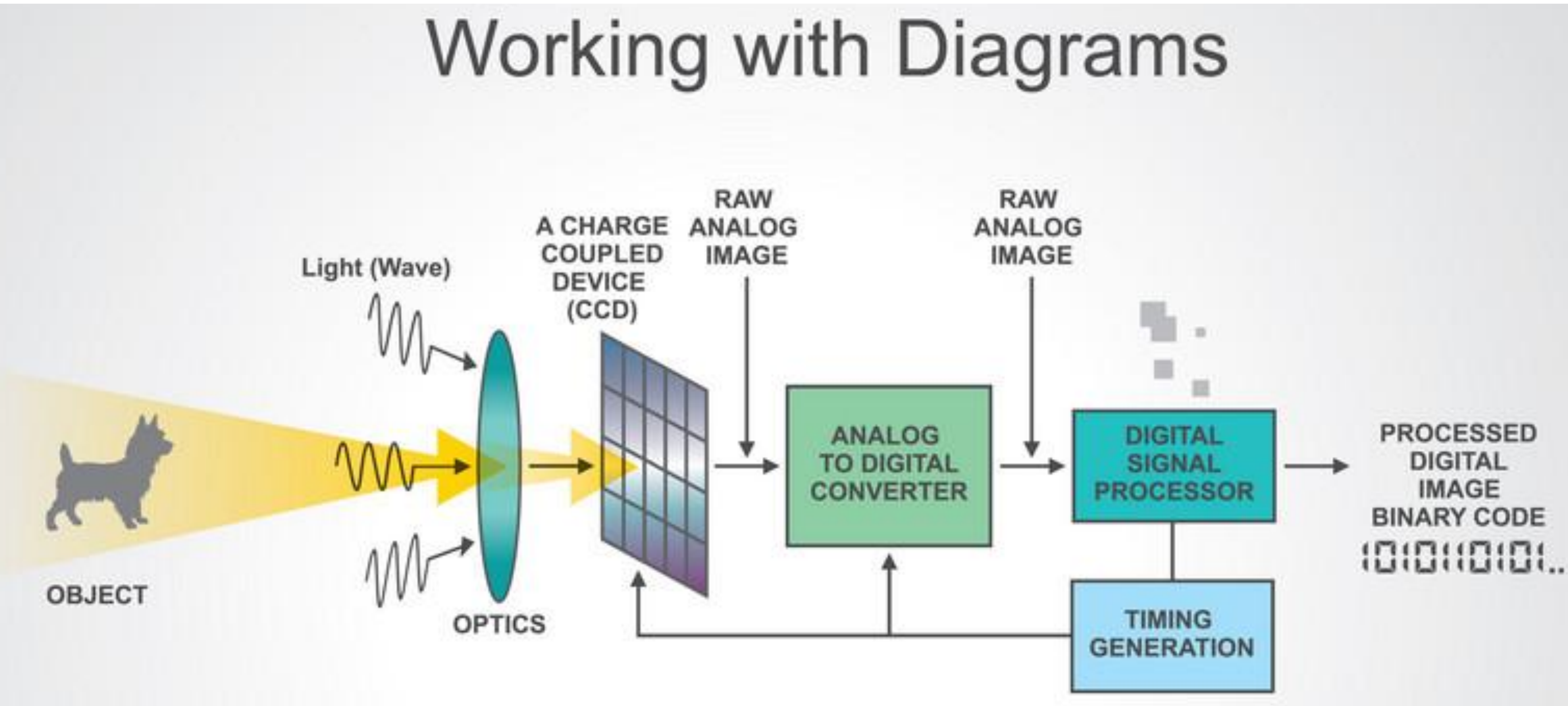
Sensors

- Light based Sensors
  - Camera
  - Infrared
  - LDR
  - X-Ray
  - UV Ray
  - Heartbeat Sensor
  - Laser Sensor
- Magnetic & Electro-Magnetic sensor
  - Reed Switch
  - Hall-Effect based Sensors
  - Electromagnetic
- Sound based Sensors (using transducer or Piezoelectricity)
  - Microphone
  - Ultra-Sonic
  - Pressure
  - Vibration

Light based Sensors

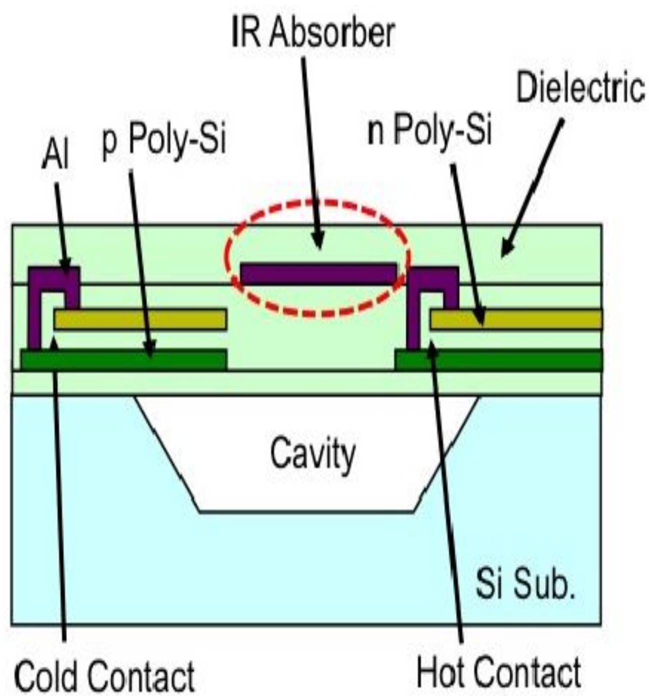
- Camera
- Infrared
- LDR
- X-Ray
- UV Ray
- Heartbeat Sensor
- Laser Sensor

Camera

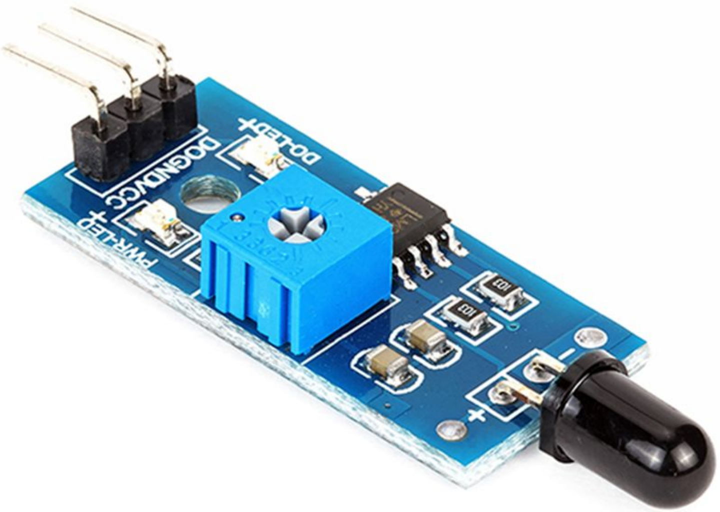

Working with Diagrams
Light (Wave)
A CHARGE COUPLED DEVICE (CCD)
RAW ANALOG IMAGE
RAW ANALOG IMAGE
OBJECT
OPTICS
ANALOG TO DIGITAL CONVERTER
DIGITAL SIGNAL PROCESSOR
PROCESSED DIGITAL IMAGE BINARY CODE
101011010101..
TIMING GENERATION

IR Sensor

Composed of Pyro electric material: Pyro electricity can be described as the ability of certain materials to generate a temporary voltage when they are heated or cooled.)

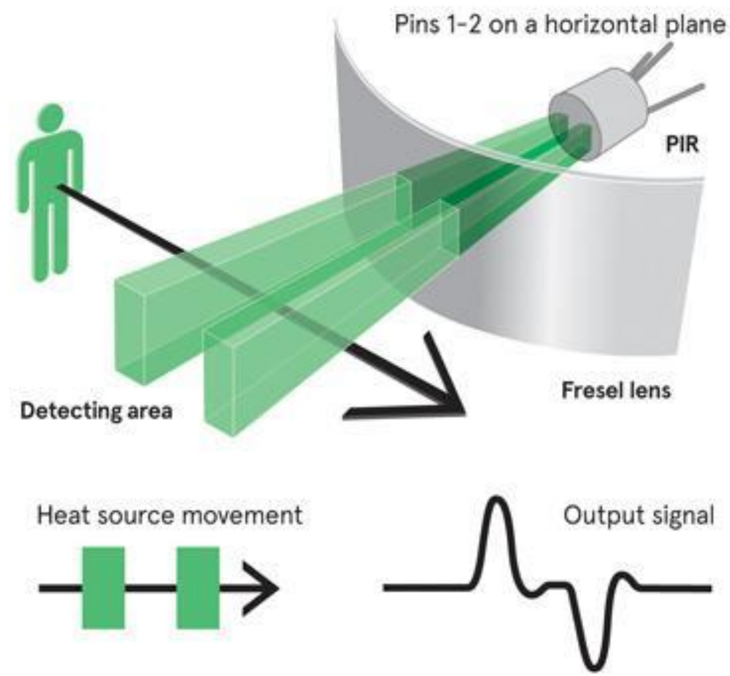


IR detector & Flame Sensor Module

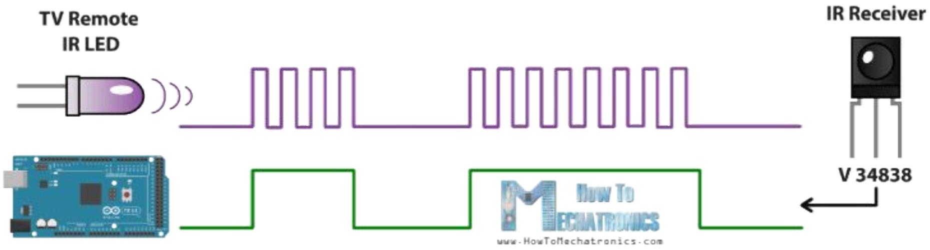


PIR

- Passive InfraRed sensor
  - Fresnel lenses used to focus the heat onto the material.
  - PIR need stabilizing period to adjust the room temperature
  - When human or animal passes heat increased and stabilized again.

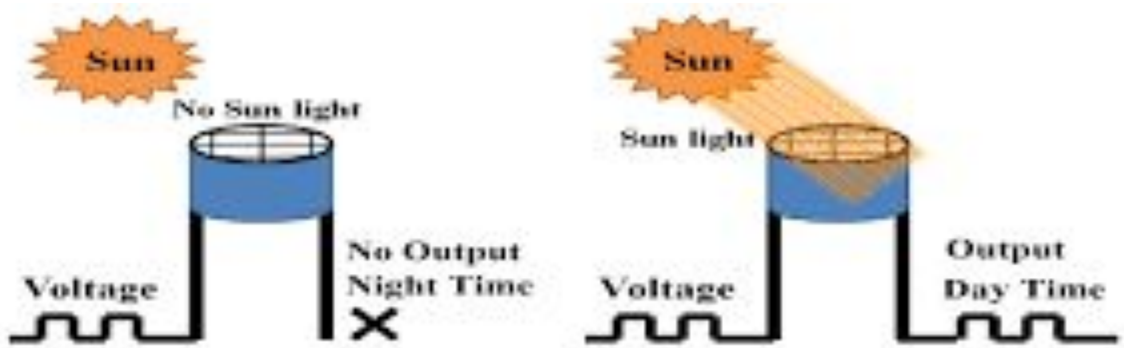


Infrared in Temperature Sensor and TV Remote

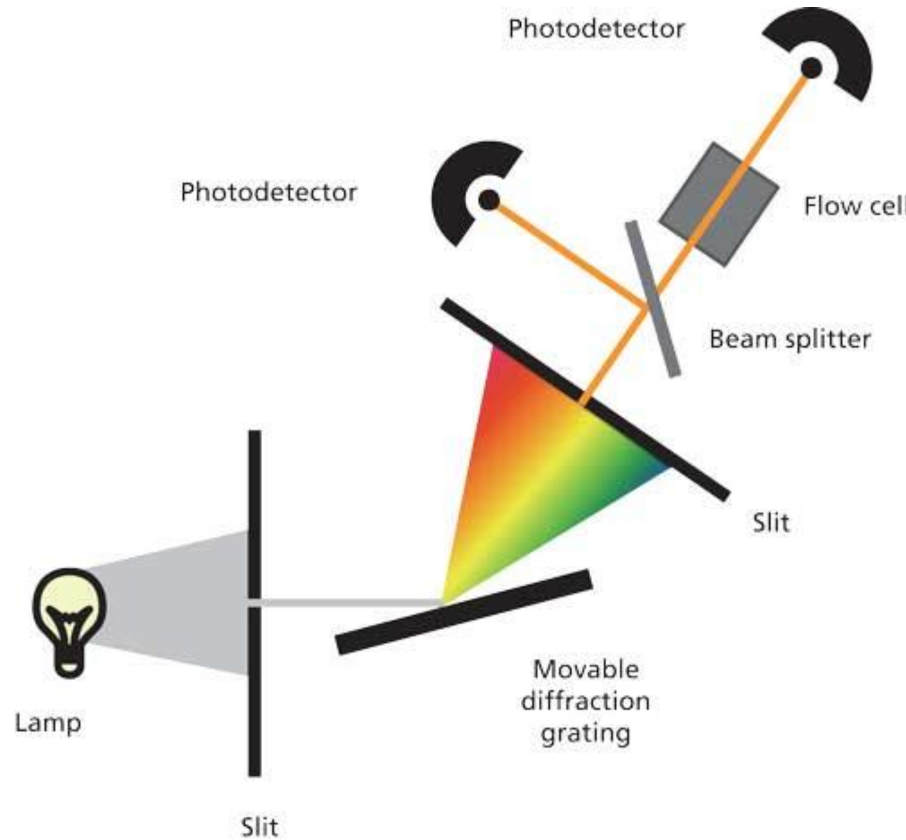


LDR

- Use Photo Sensitive Semi-conductor material



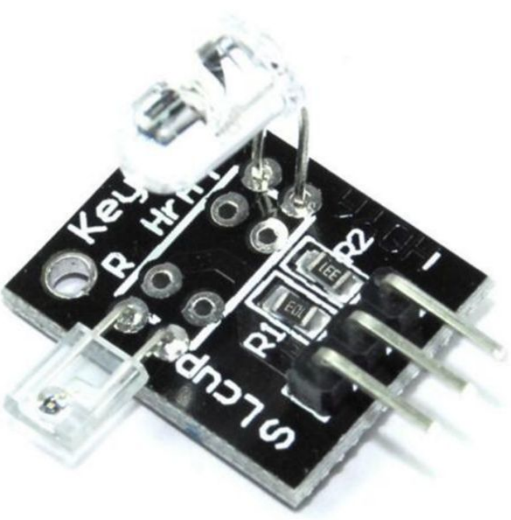
UV Sensor Module

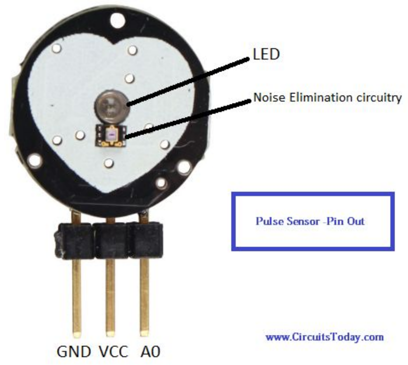


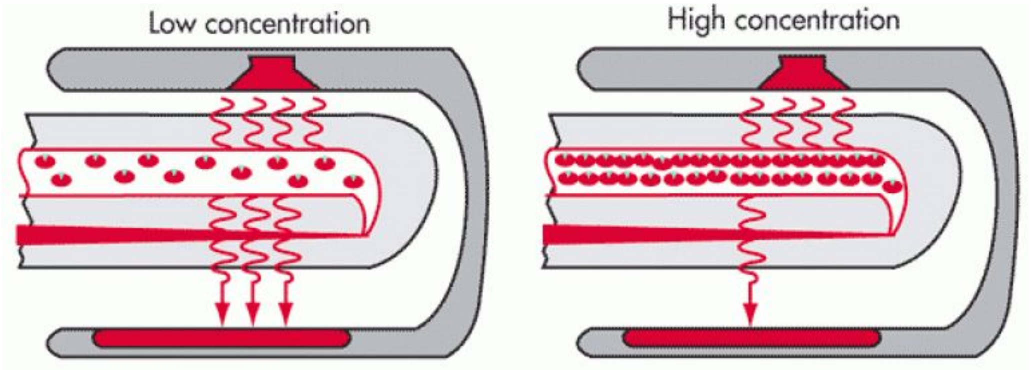
Response wavelength: 200~370nm

X-Ray Sensor

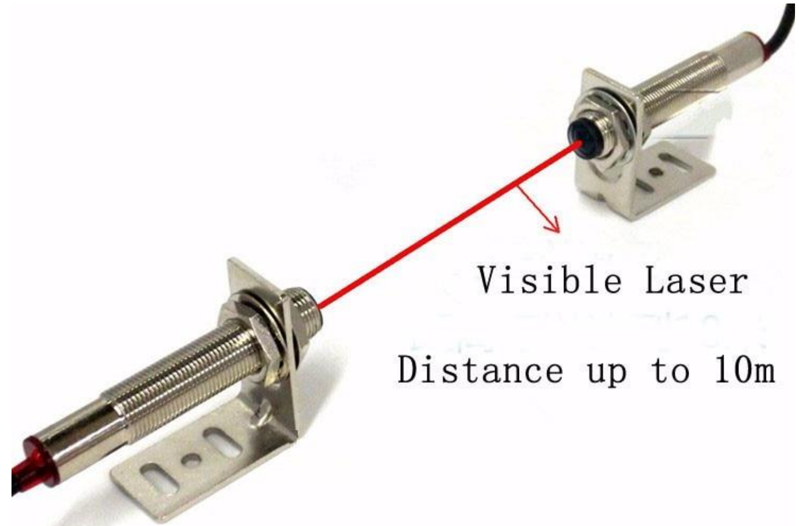
Pulse Oximeter using light and IR

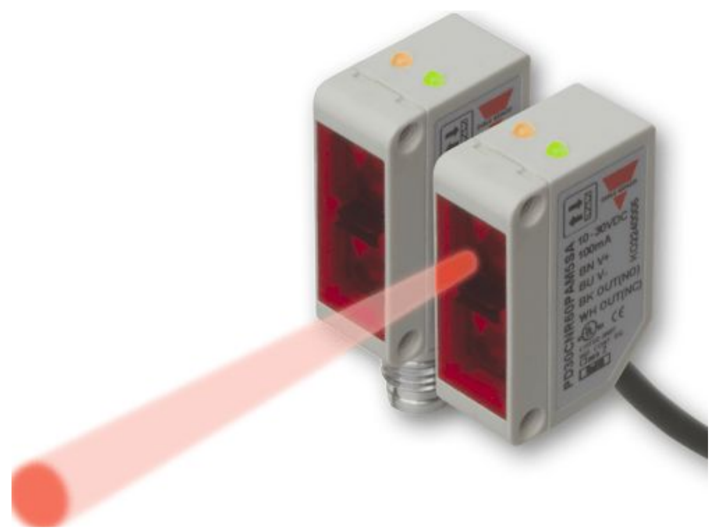






Laser Sensor

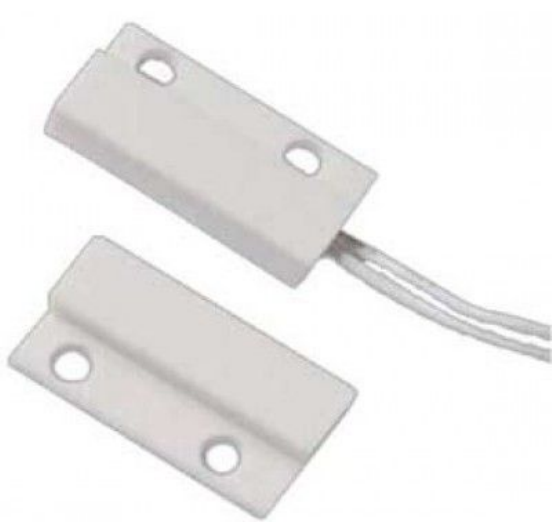


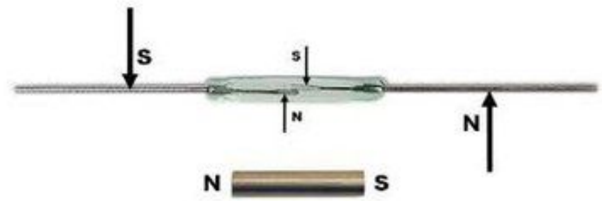


Magnetic & Electro-Magnetic sensor

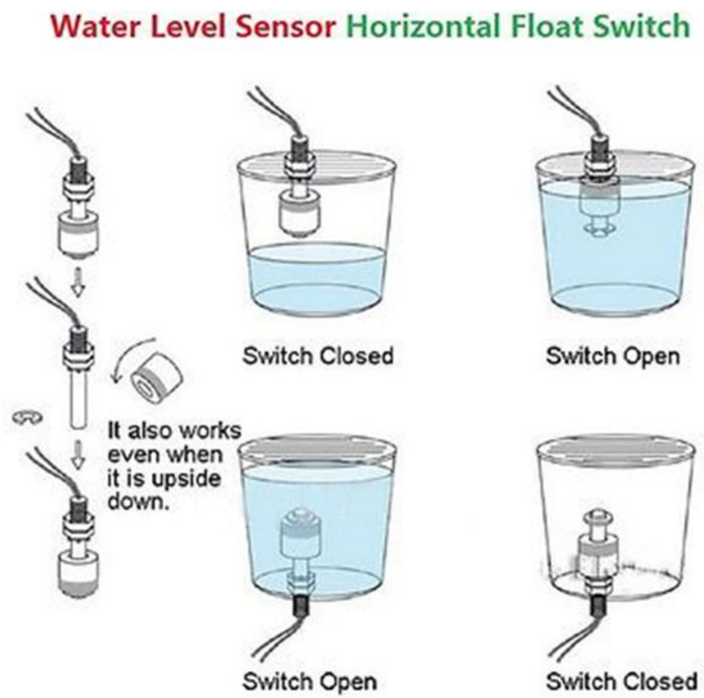
- Reed Switch
- Hall-Effect based Sensors
- Electro-Magnetic Sensor

Magnetic Type Sensor
(Mini Magnetic or Reed Switch)

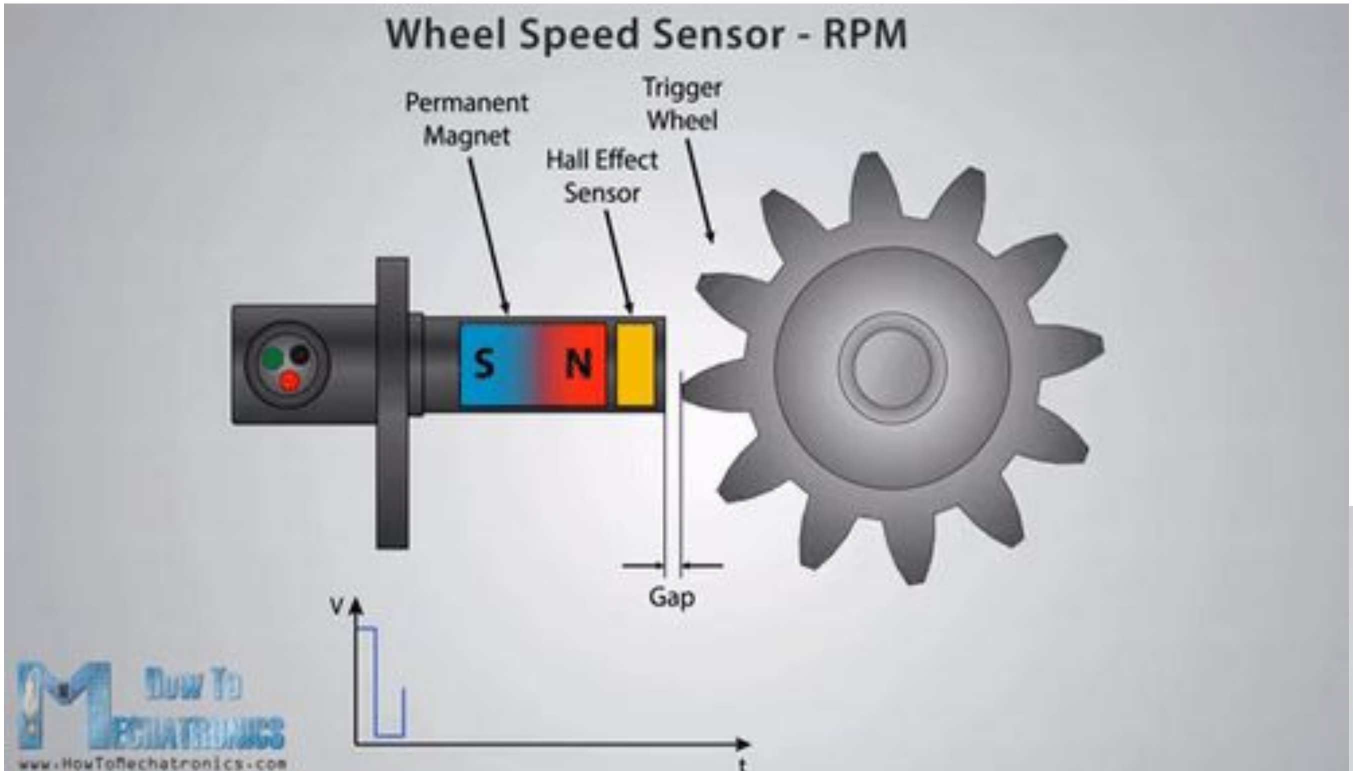




Liquid Level Sensor using reed switch



Hall Effect



Wheel Speed Sensor - RPM
Permanent Magnet
Trigger Wheel
Hall Effect Sensor
S
N
Gap
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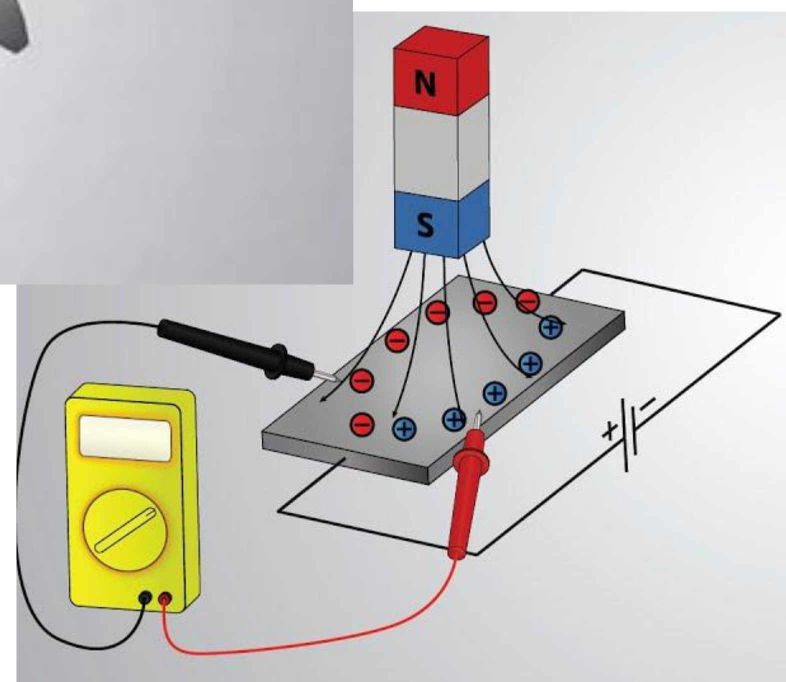
www.HowToMechatronics.com



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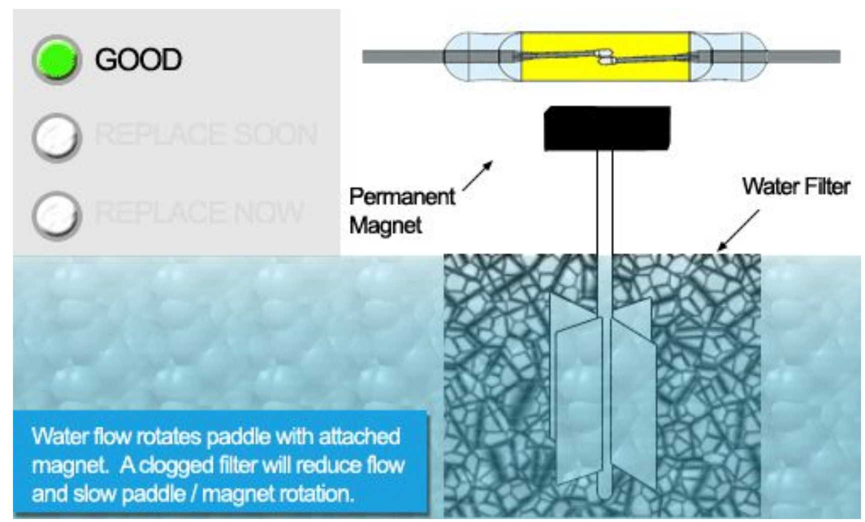
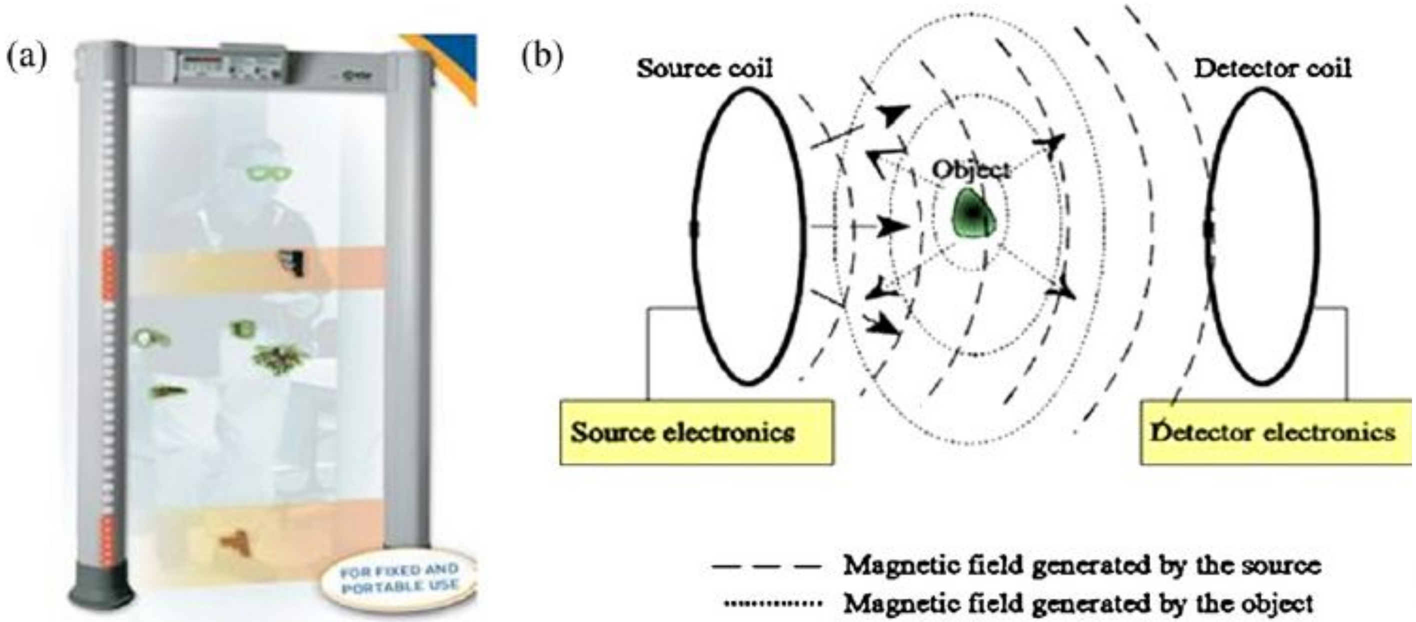
Water Flow Sensor using both reed switch and Hall effect



Metal Detector using Hall-effect



Sound based Sensor

- By using Transducer or Magnetic Flux
- By using Piezoelectric

- Microphone
- Ultra-Sonic
- Pressure
- Vibration

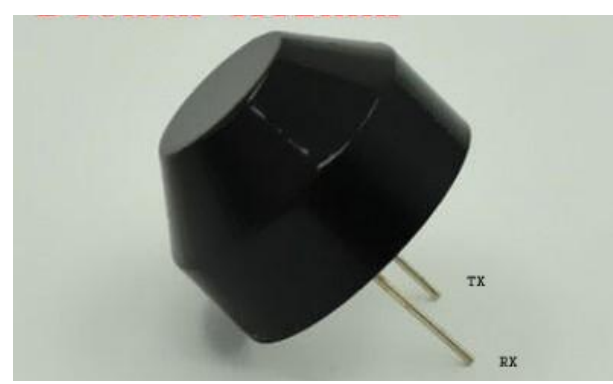
Piezoelectric Materials

- Natural piezoelectric occurring materials:
  - Berlinite (structurally identical to quartz)
  - Cane sugar
  - Quartz (SiO_2)
  - Rochelle salt
  - Topaz
  - Tourmalin
  - Dry bone
- Man-made piezoelectric materials:
  - Barium titanate
  - Lead zirconate titanate.

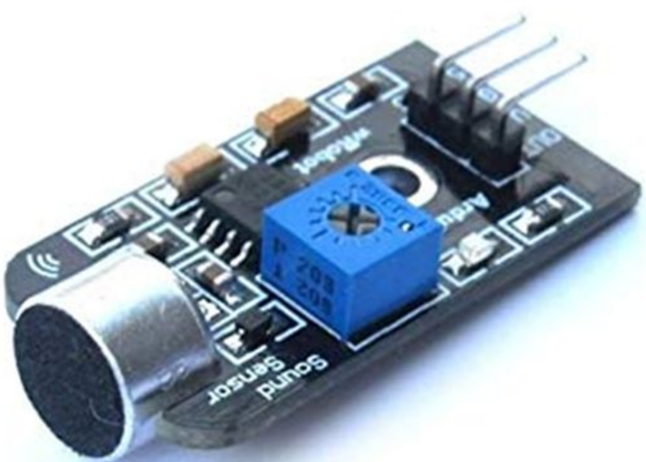
Ultrasonic

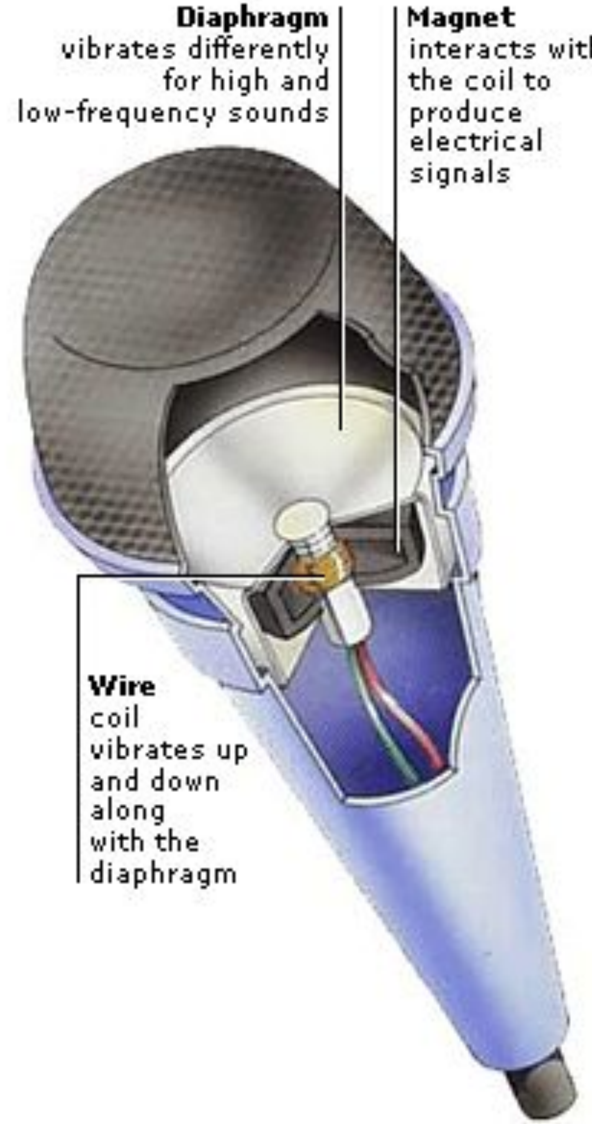
- By using travelling time of a ultrasonic wave with frequency of 40khz

- **Distance = (Speed X Time)/2**
- **Speed of ultrasonic wave is: 340meter/second in air**

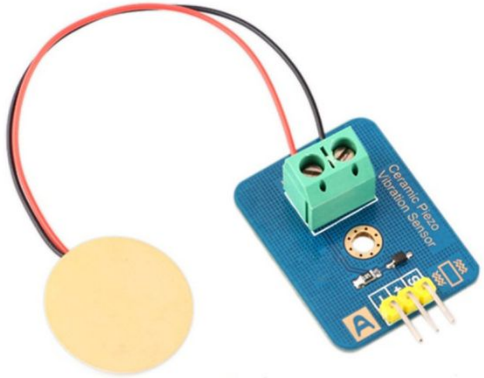


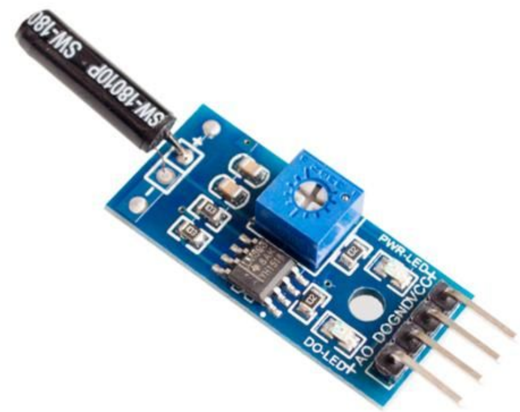
SOUND SENSOR (microphone)

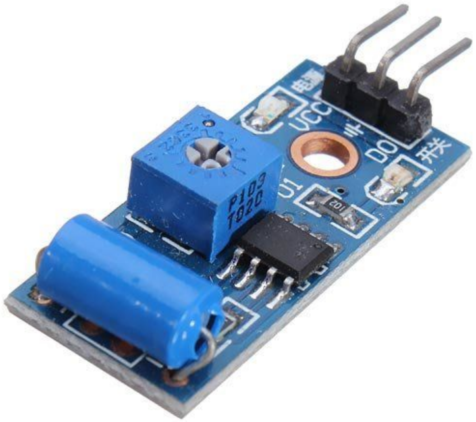




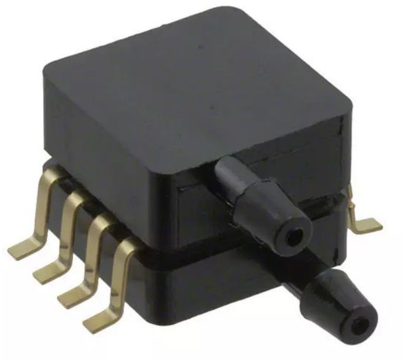
Vibration Sensor (Using Piezo)







Pressure Sensor



- Pressure of gas and liquid.

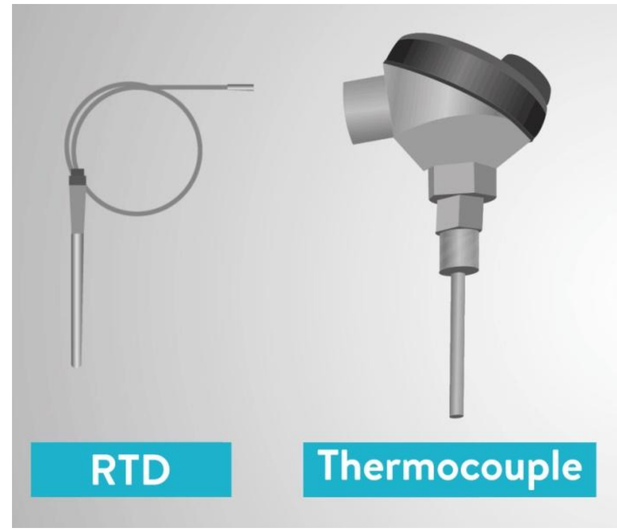
Modern Sensors

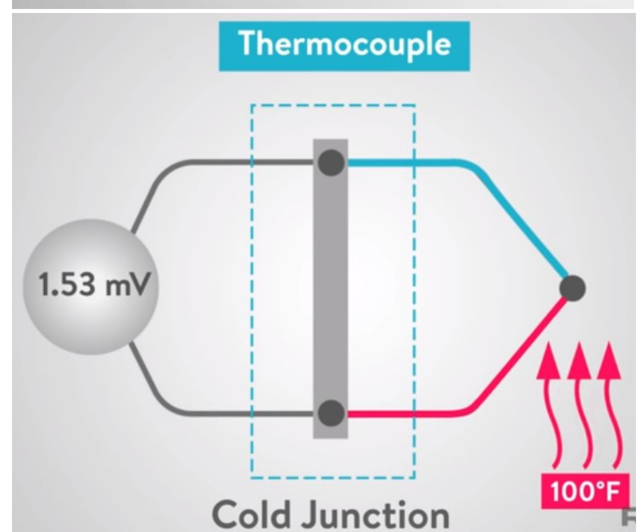
- Temperature
- Proximity
- Smoke & Gas
- Voltage Sensor
- Humidity Sensor
- Touch Sensor
- Gyroscope
- Accelerometer
- Break-Beam
- LiDAR

- https://www.youtube.com/watch?v=xvF8b3eKelU
- https://youtu.be/4mQ3o1t4Ssg
- https://www.youtube.com/watch?time_continue=2&v=4mQ3o1t4Ssg&feature=emb_logo

Temperature Sensing

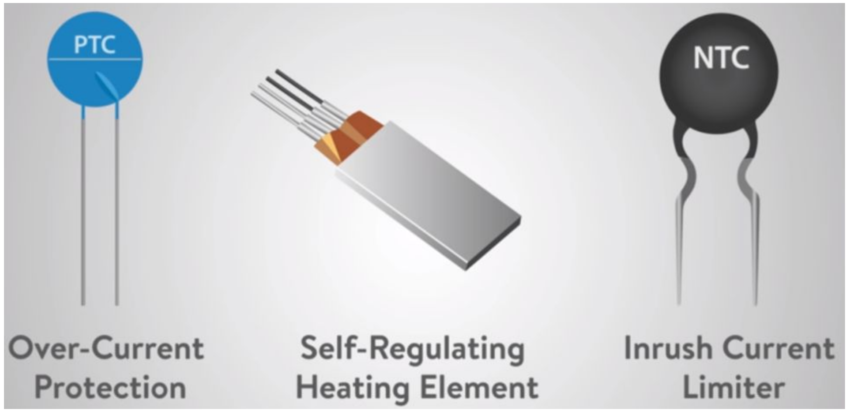
- Resistance Temperature Detector (RTD): Platinum, Copper, Nickel, Nickel-Ion
- Thermocouple (TC): Junction of two materials (Iron-constantan, Cromel-Alimel)

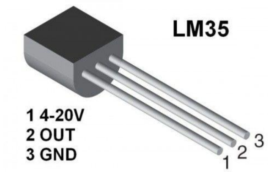




Temperature Sensing

- Thermistor are temperature dependent resister
  - PTC (Positive Temperature Coefficient)
  - Self regulating Heating Element
  - Negative Temperature Coefficient NTC
- Semiconductor based temperature sensor used inside IC





RTD resistors are pure metal and thermistors are made with polymer or ceramic materials

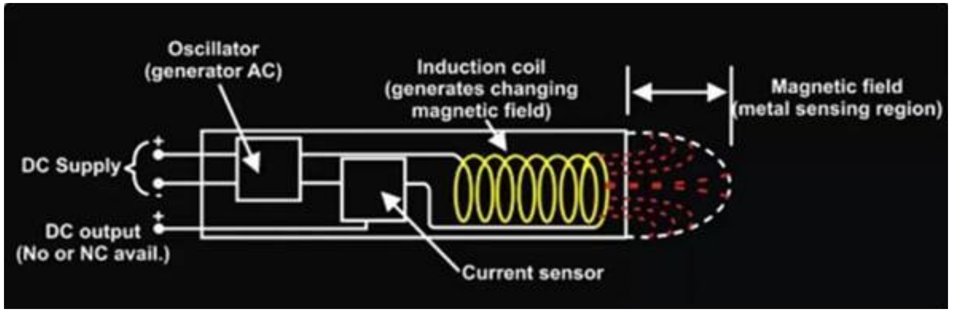
Types of Proximity Sensors

1. **Inductive Sensors**
2. **Capacitive Sensors**
3. **Photoelectric Sensor**
4. **Magnetic Sensors**

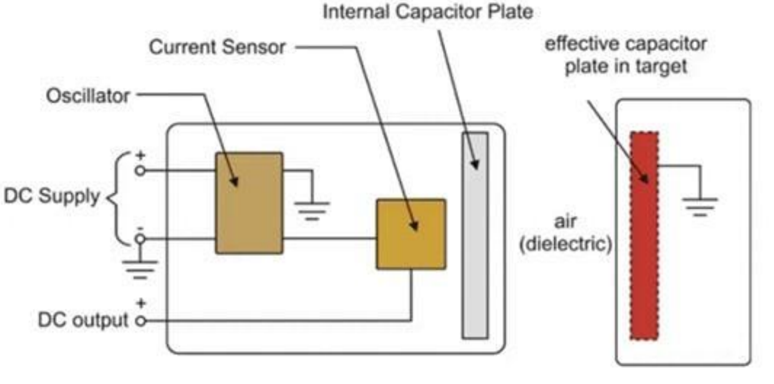
https://www.youtube.com/watch?v=xAiSuQK22IM

Proximity Sensors

- Inductive:
  - Measure metal magnetic field from induction coil

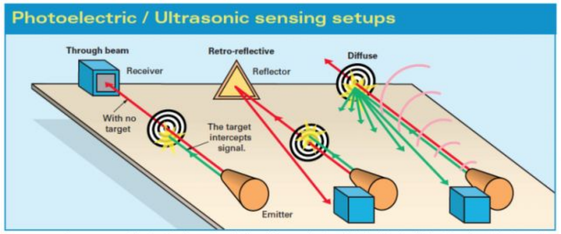


- Capacitive:
  - Measures metallic and non-metallic objects (Liquids, plastics, woods etc.)



Other ways of proximity sensing

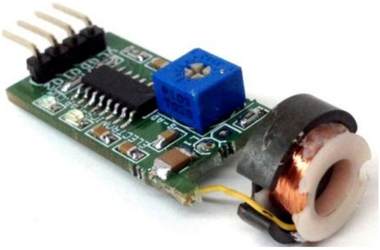
- Photoelectric:



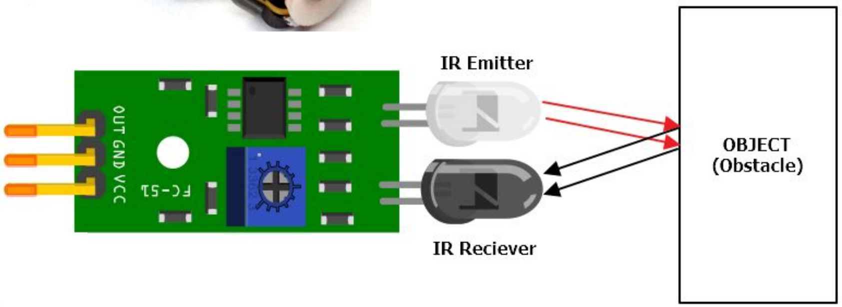
Lasers or sound waves serve as the signal in three setups. In through beam and retro-reflective, the signal shoots from the emitter to receiver until the target cuts it off. In diffuse sensing, the signal diverges until a target moves in and reflects some back to the receiver.

- Magnetic:

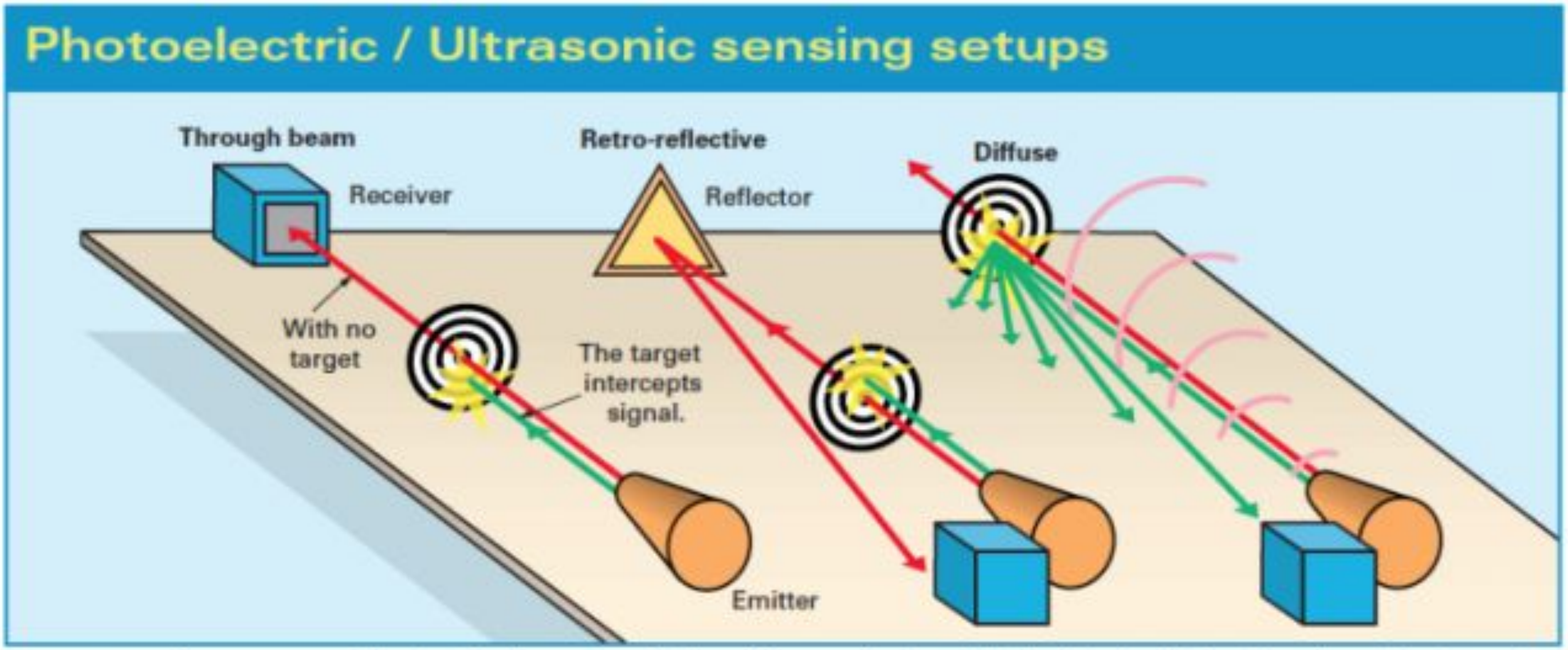
- Inductive:



- IR

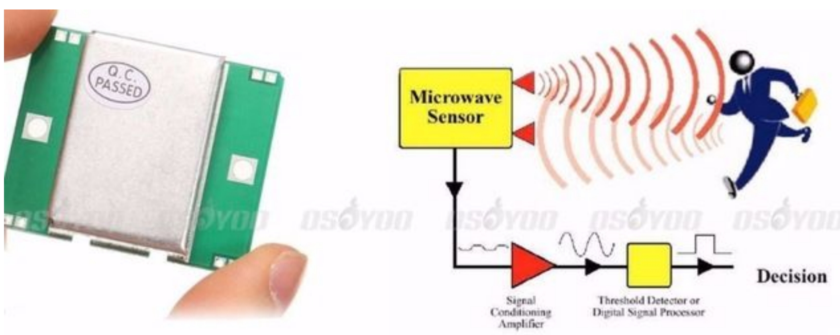


- Laser

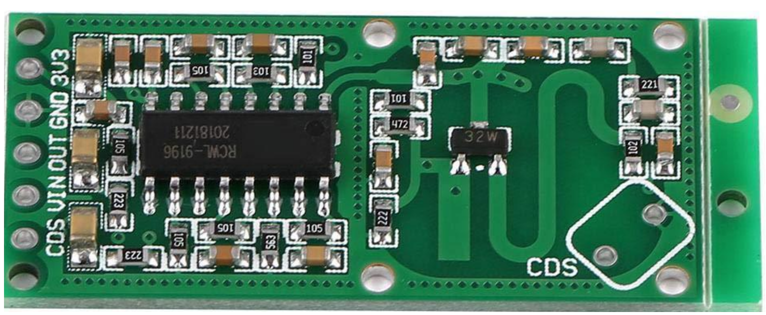


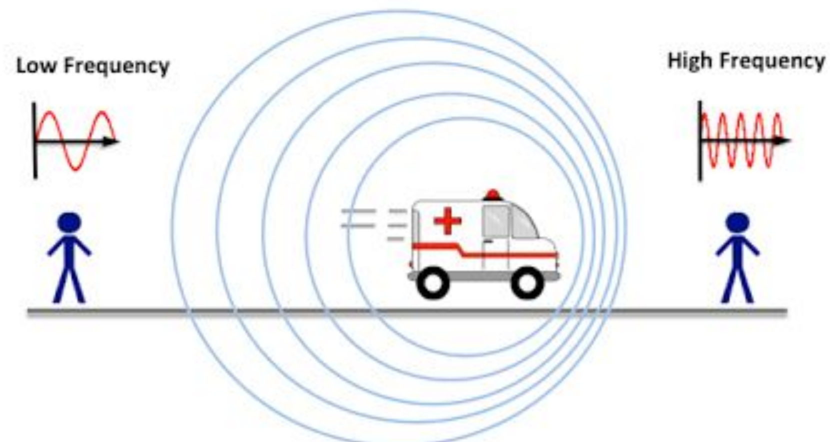
Lasers or sound waves serve as the signal in three setups. In through beam and retro-reflective, the signal shoots from the emitter to receiver until the target cuts it off. In diffuse sensing, the signal diverges until a target moves in and reflects some back to the receiver.

Microwave Proximity Sensor (for moving object)



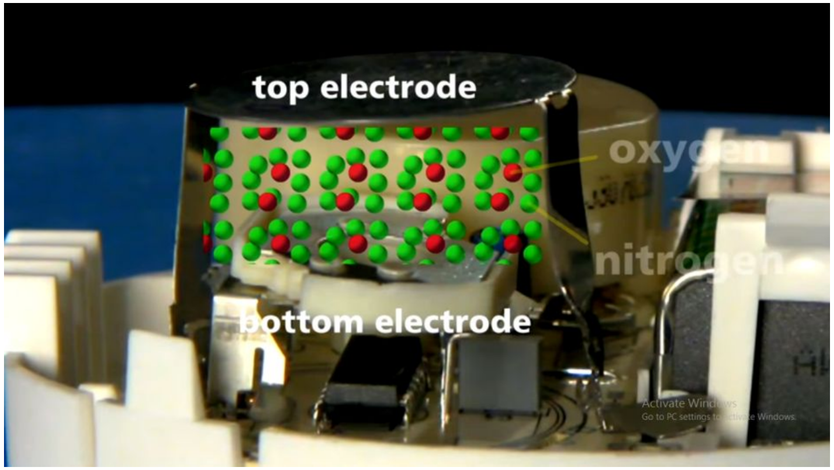
- Using “Doppler Effect” of moving object





Smoke Detector

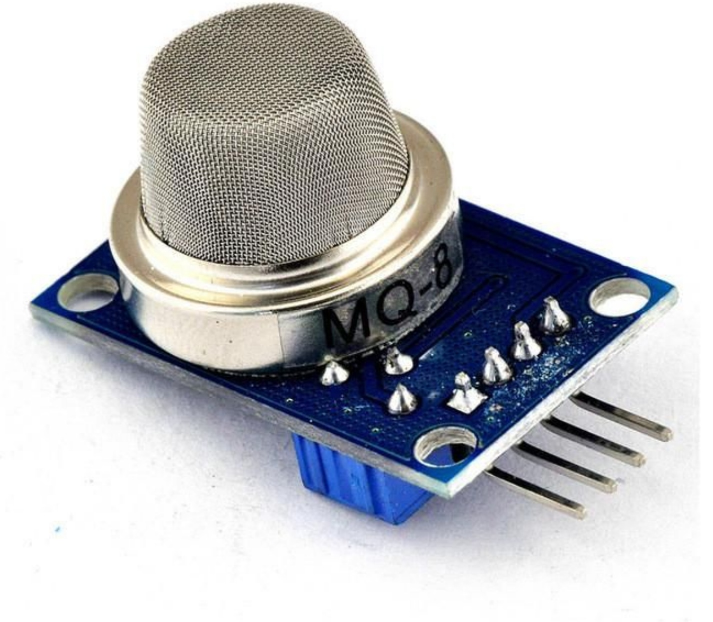
- Radio active material: Americium-241



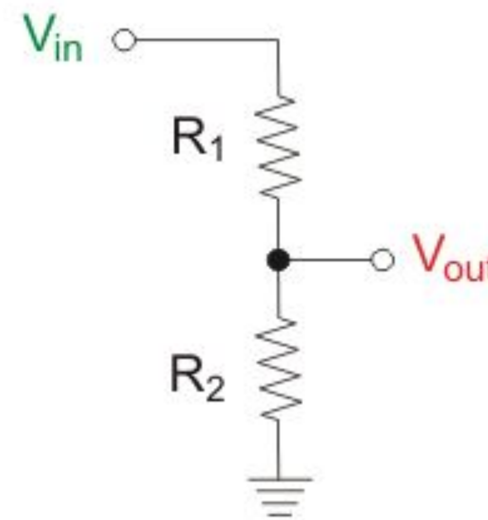
Smoke and Gas Sensor

Materials: metal oxides, semiconductors, carbon-based materials, and conducting polymers etc. Example: (Cami resistance / Carbon nano-tube)

1. Gas Sensor (MQ-9)
2. Smoke Sensor (MQ-5)
3. MQ135 Air Quality Sensor module
4. MQ6 LPG, LNG, iso-butane, propane gas
5. MQ8 Hydrogen gas sensor Module

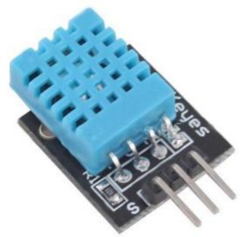


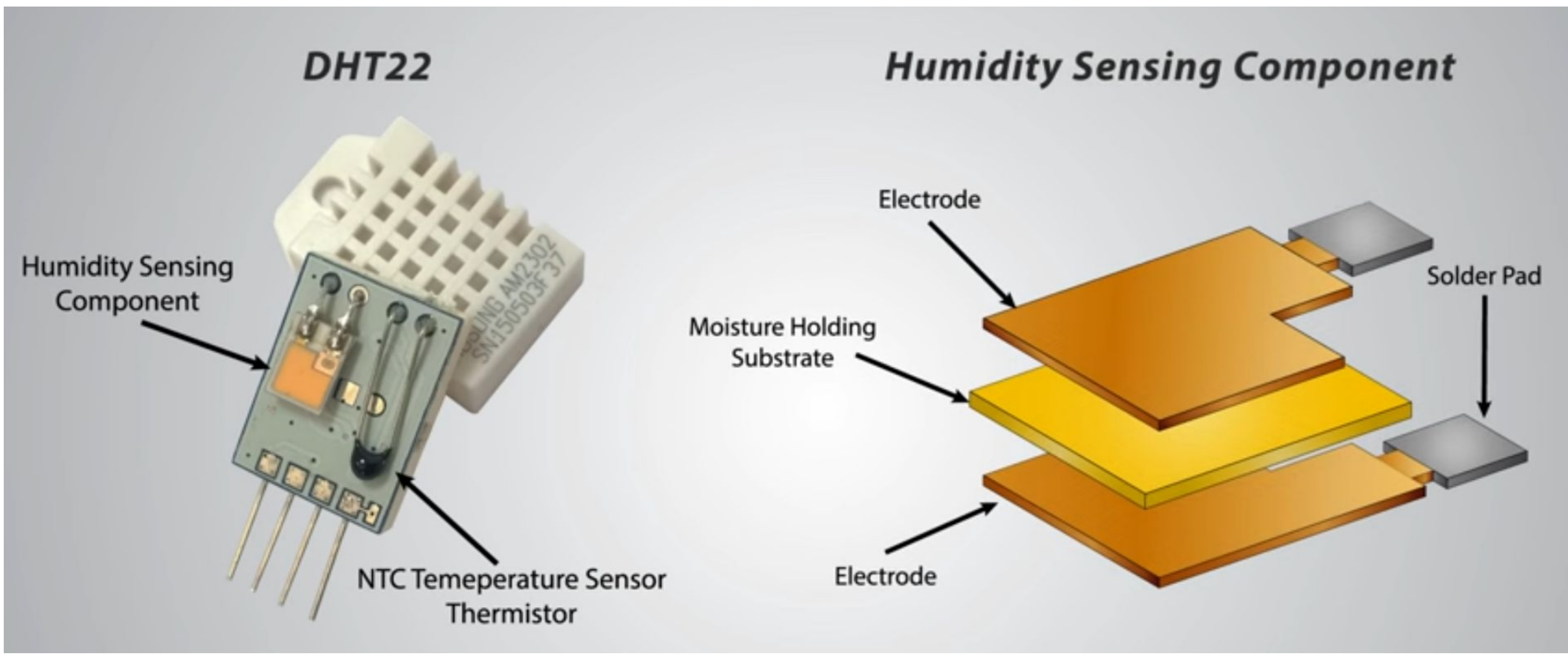
Voltage Sensor



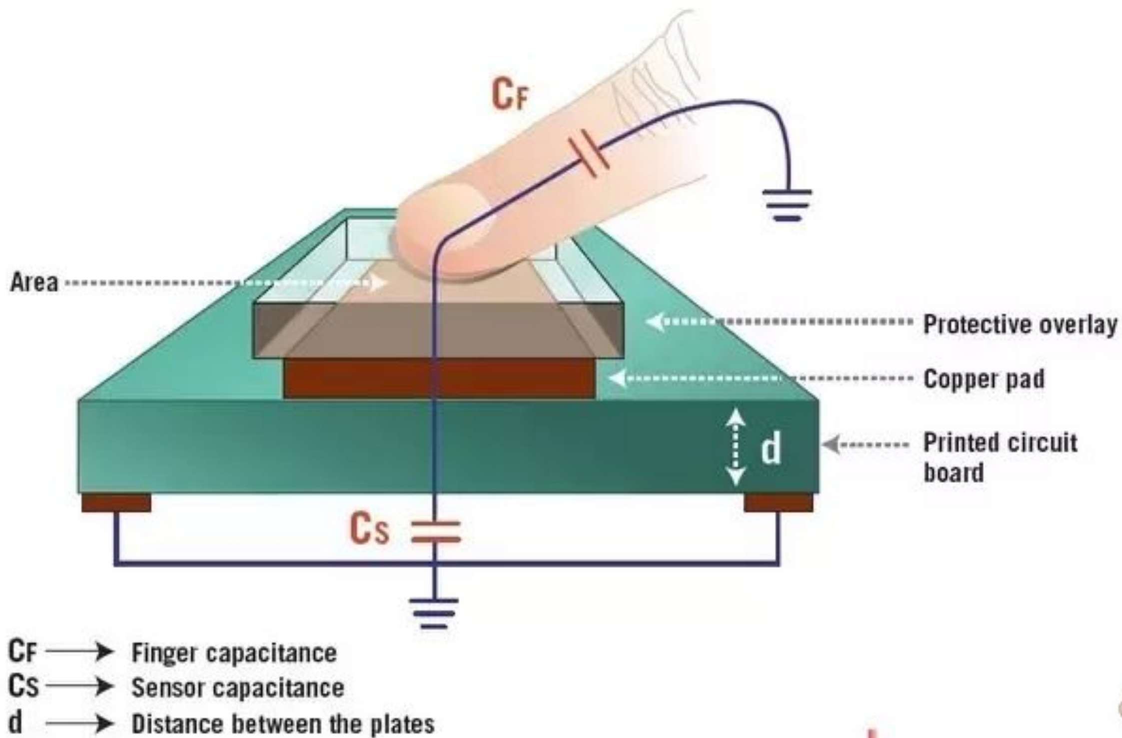
$$V_{out} = \frac{R_2}{R_1 + R_2} \times V_{in}$$

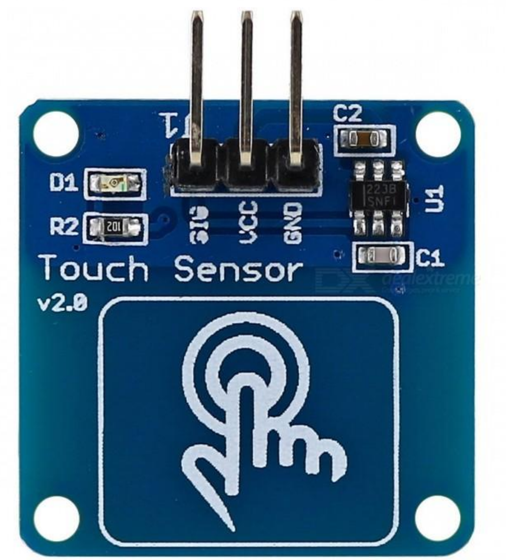
Humidity Sensor

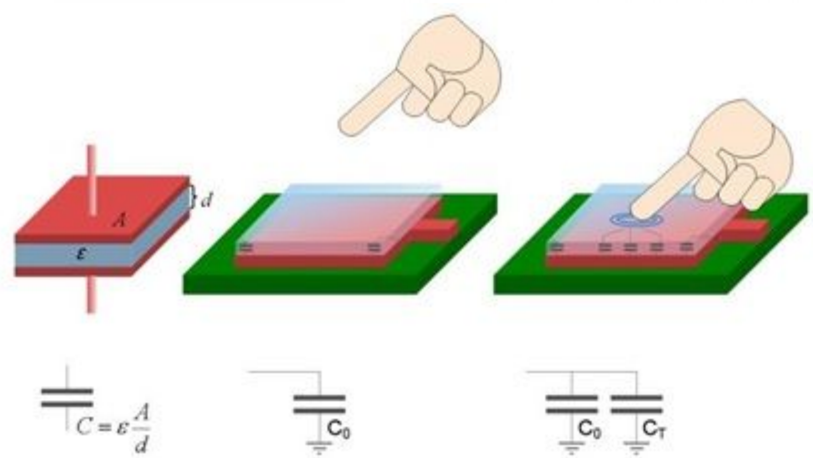




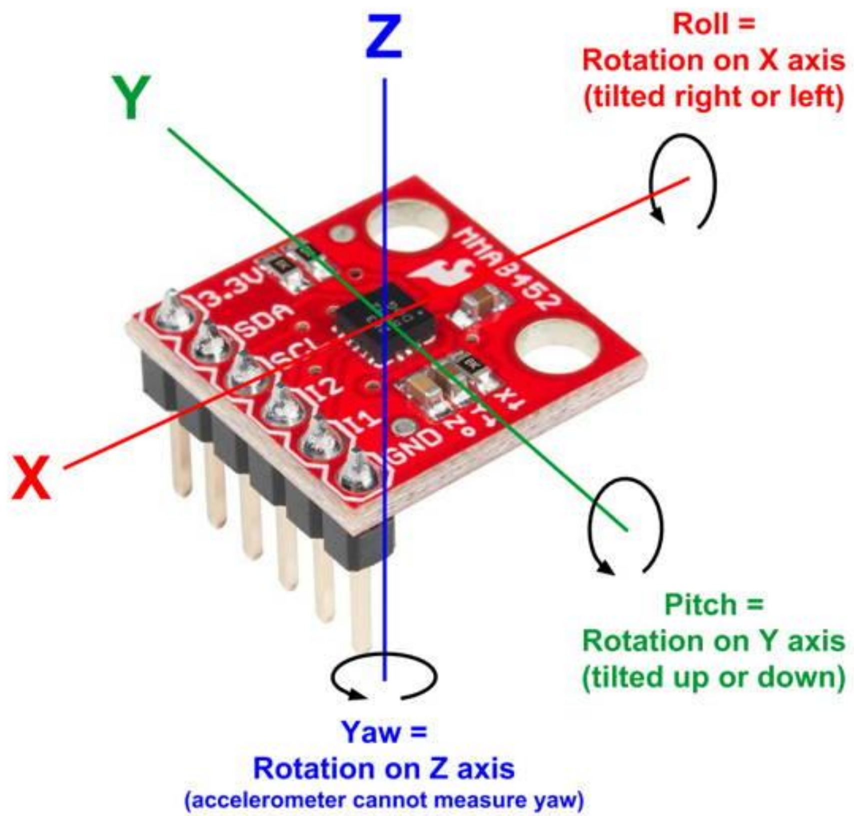
Touch Sensor



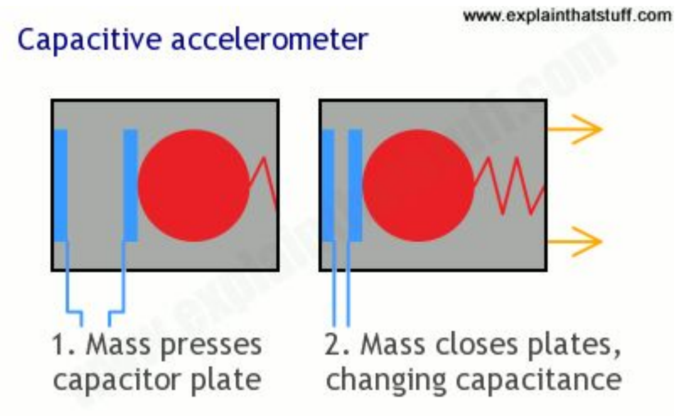


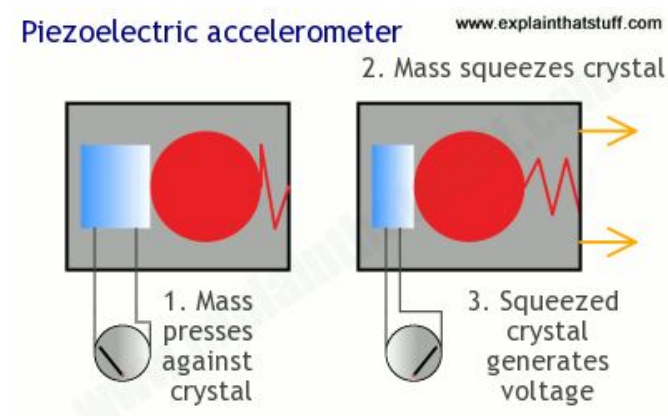


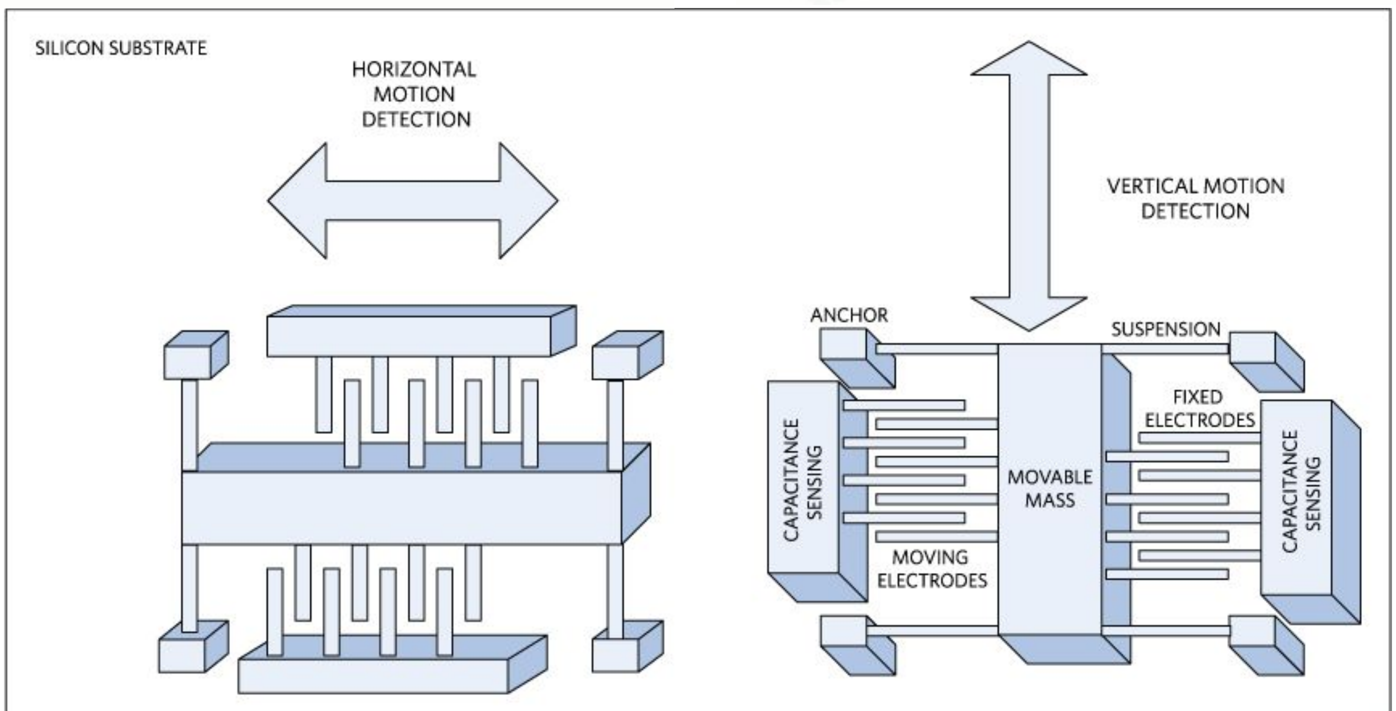
Accelerometer



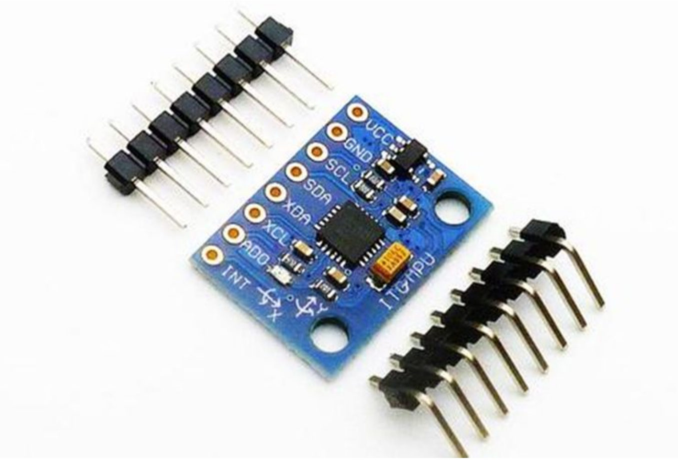
Accelerometer

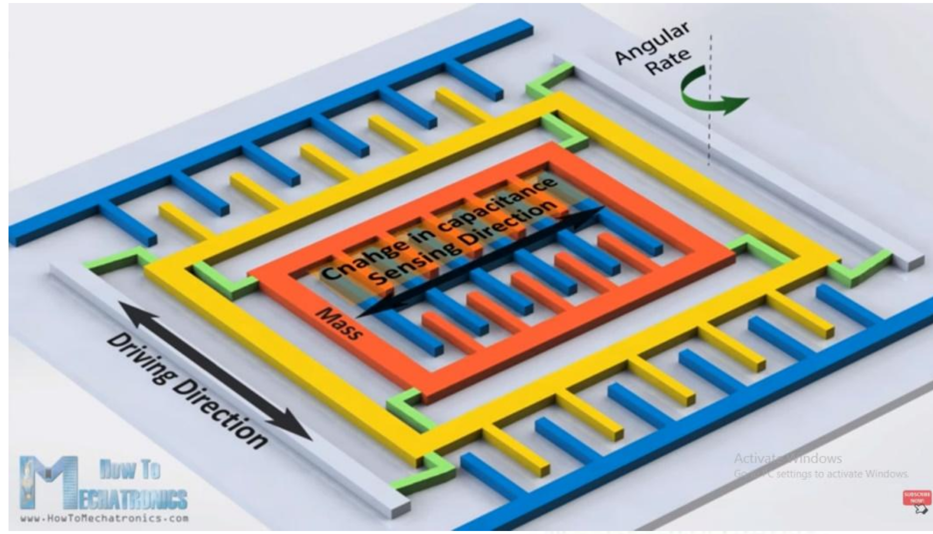






Gyroscope


VCC
GND
SCL
SDA
XDA
XCL
ADO
INT
ITG/MPU


Angular Rate
Chahge in capacitance Sensing Direction
Mass
Driving Direction
How To Mechatronics
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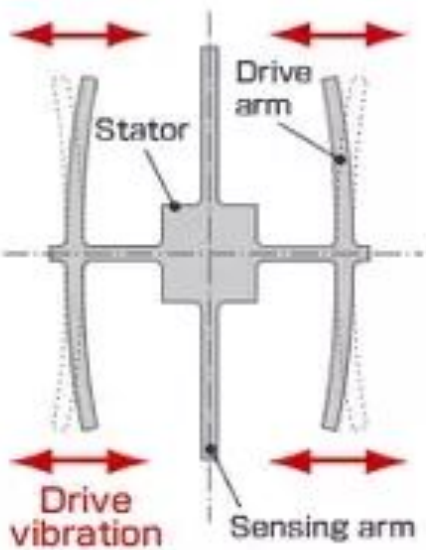
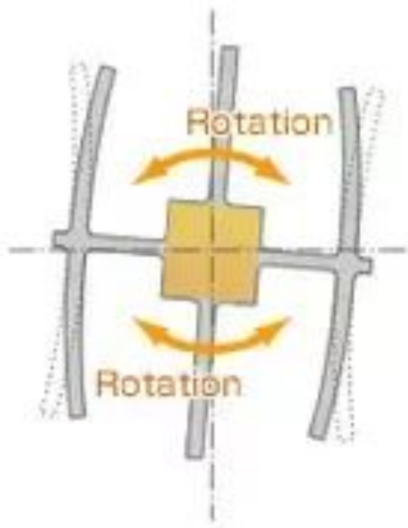
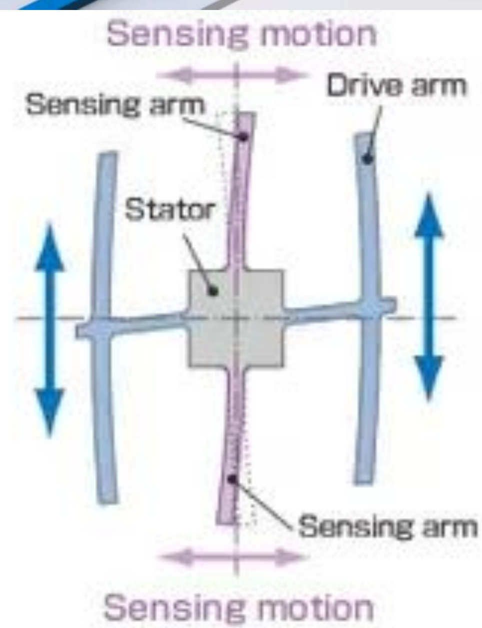
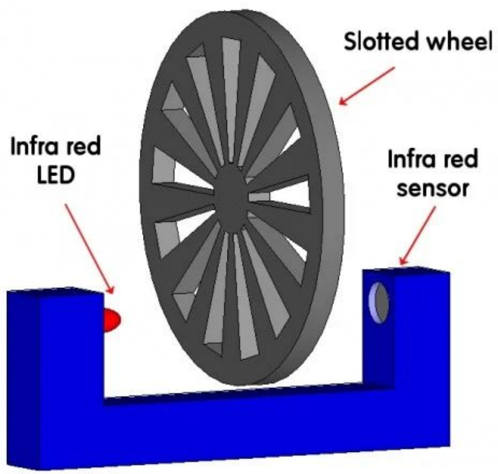
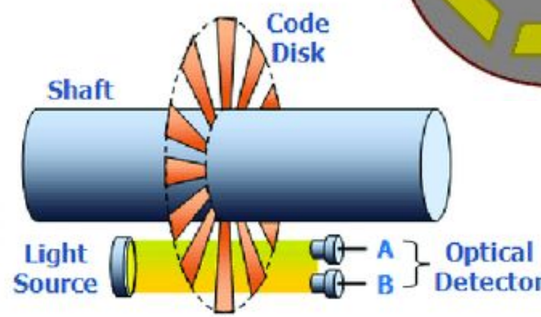

Stator
Drive arm
Drive vibration
Sensing arm

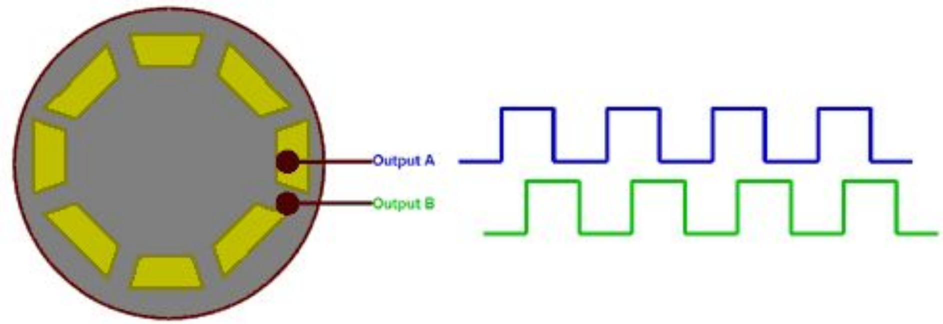

Rotation
Rotation

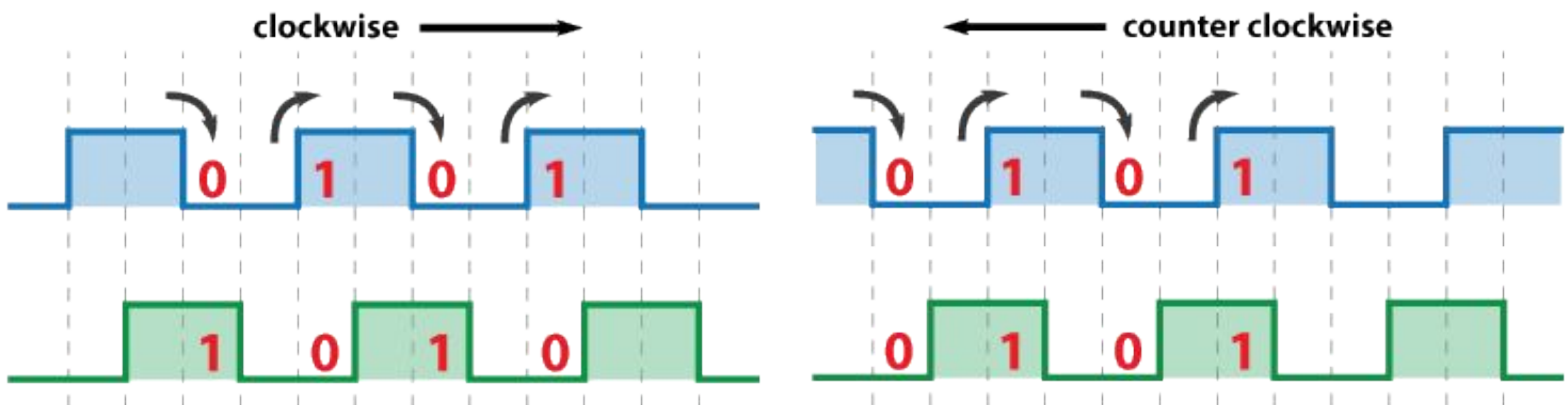

Sensing motion
Drive arm
Sensing arm
Stator
Sensing arm
Sensing motion

Break beam sensor (Shaft/Rotary encoder)

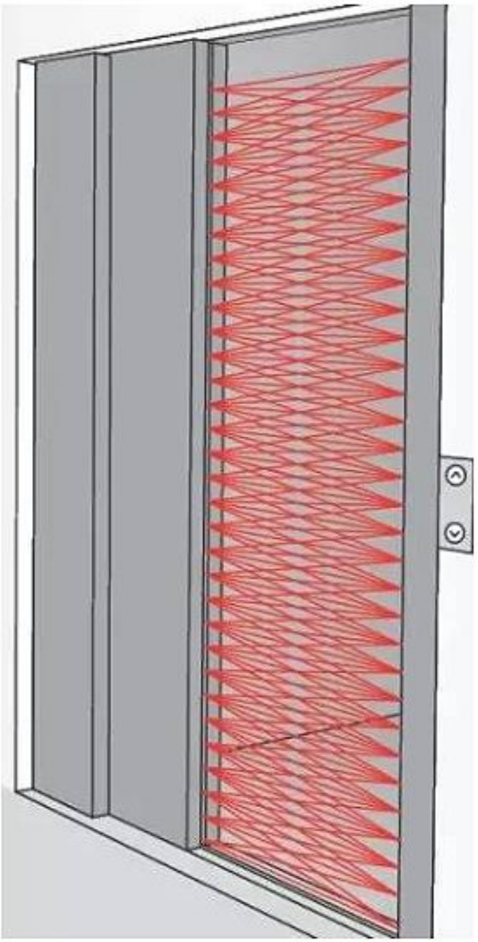


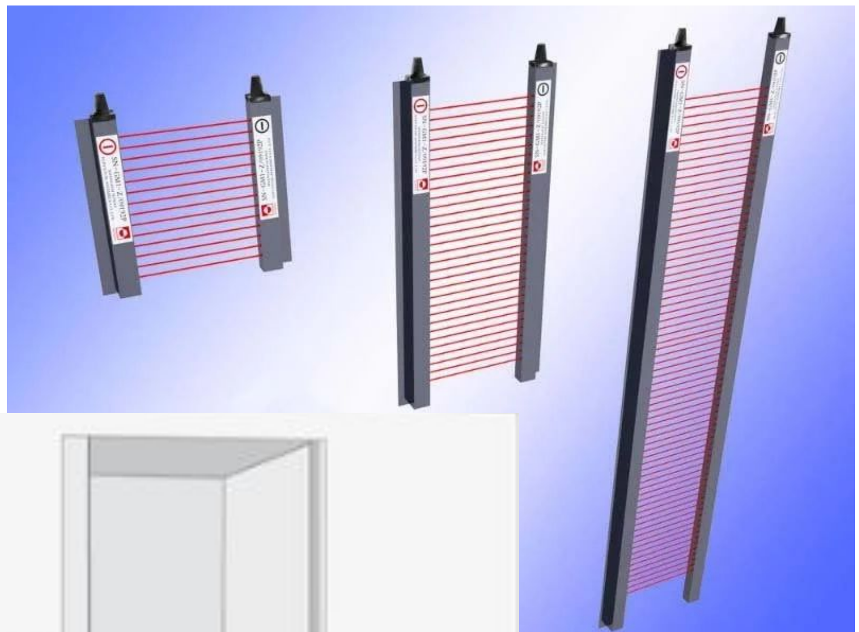


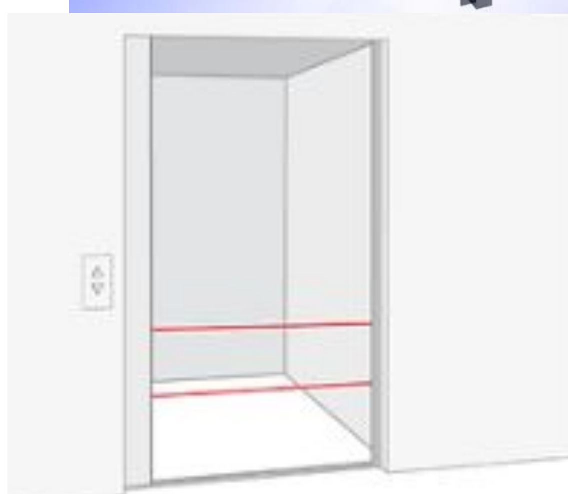




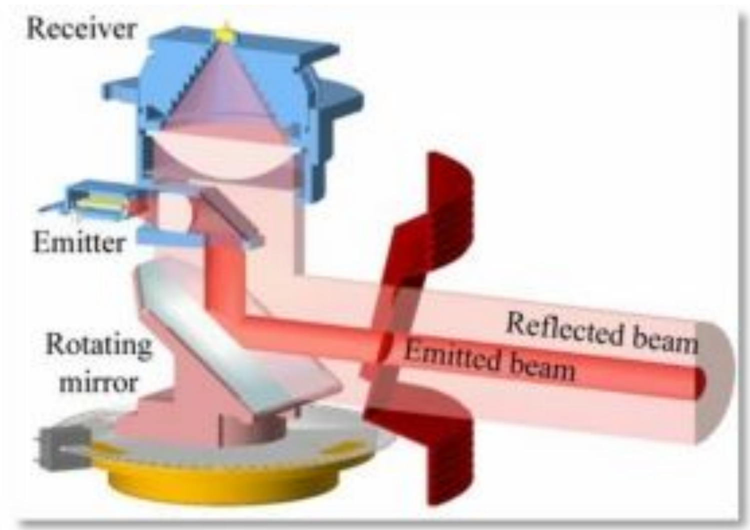
Break beam sensor
(Light curtain (Lift door))





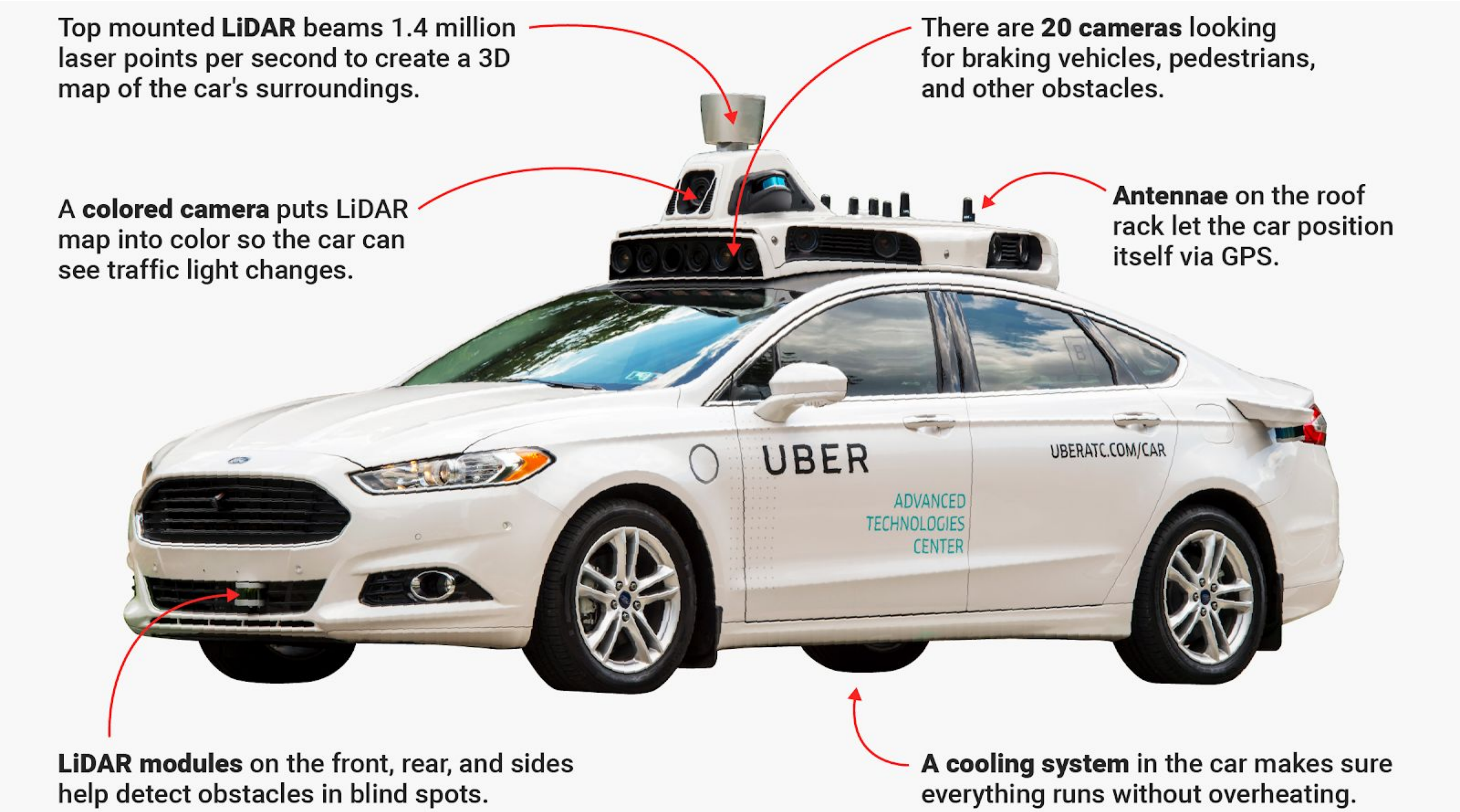


LiDAR (Light Detection And Ranging)



Velodyne LiDAR

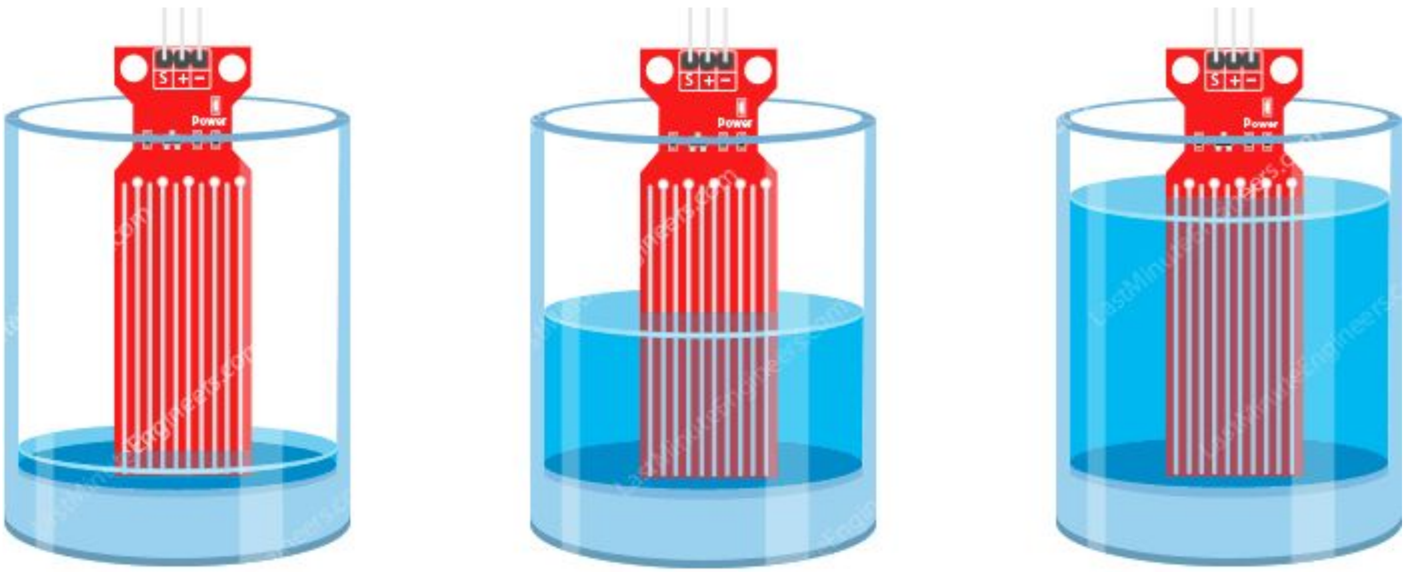
Laser Scanner or Lidar



Water and Underwater Sensors

- Water Conductivity based sensor
  - Steam
  - Rain
  - water level
  - Soil Moisture Sensor
- Underwater Sensor:
  - Ph and Alkalinity
  - Dissolved Oxygen Sensor (DO)
  - Biochemical Oxygen Demand (BOD)
  - BOD, COD, TSS
  - Total Dissolved Solids S and Turbidity
  - Chlorine
  - Electrical Conductivity sensor water

Water Level sensor

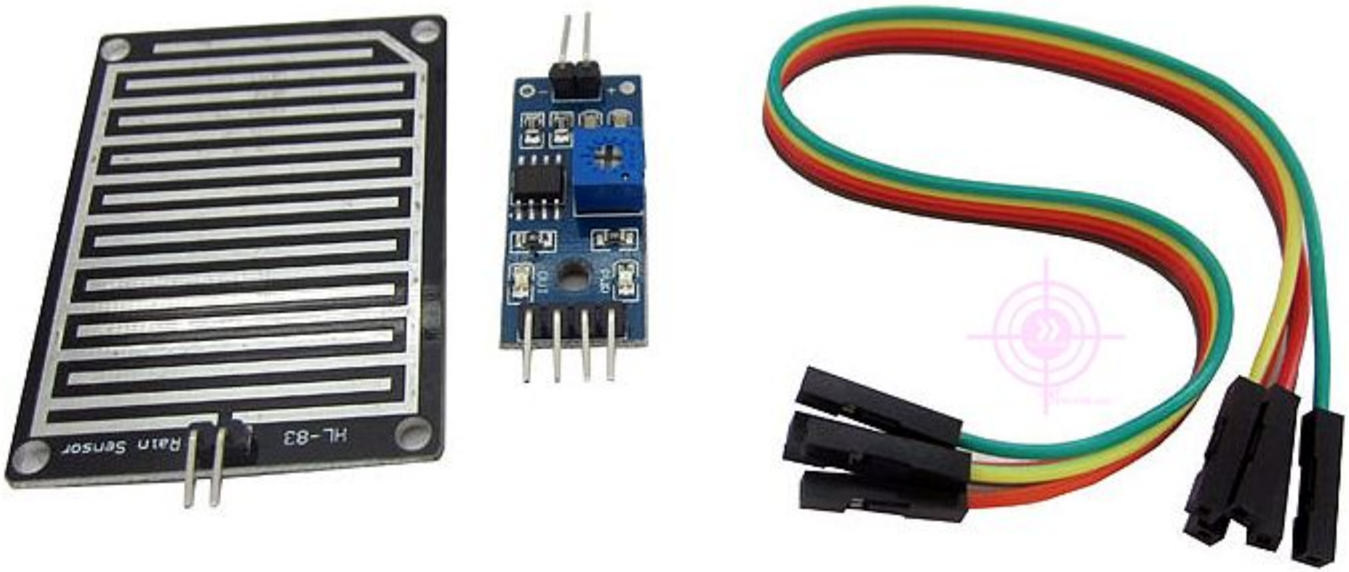


Status: Dry
Test Reading: ~0

Status: Partially submerged
Test Reading: ~420

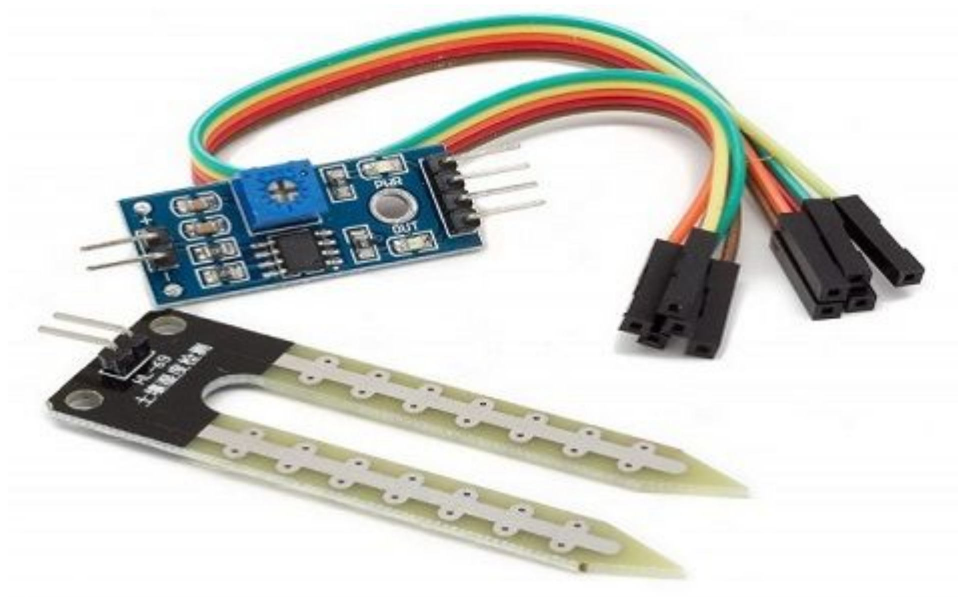
Status: Fully submerged
Test Reading: ~520

Rain Sensor



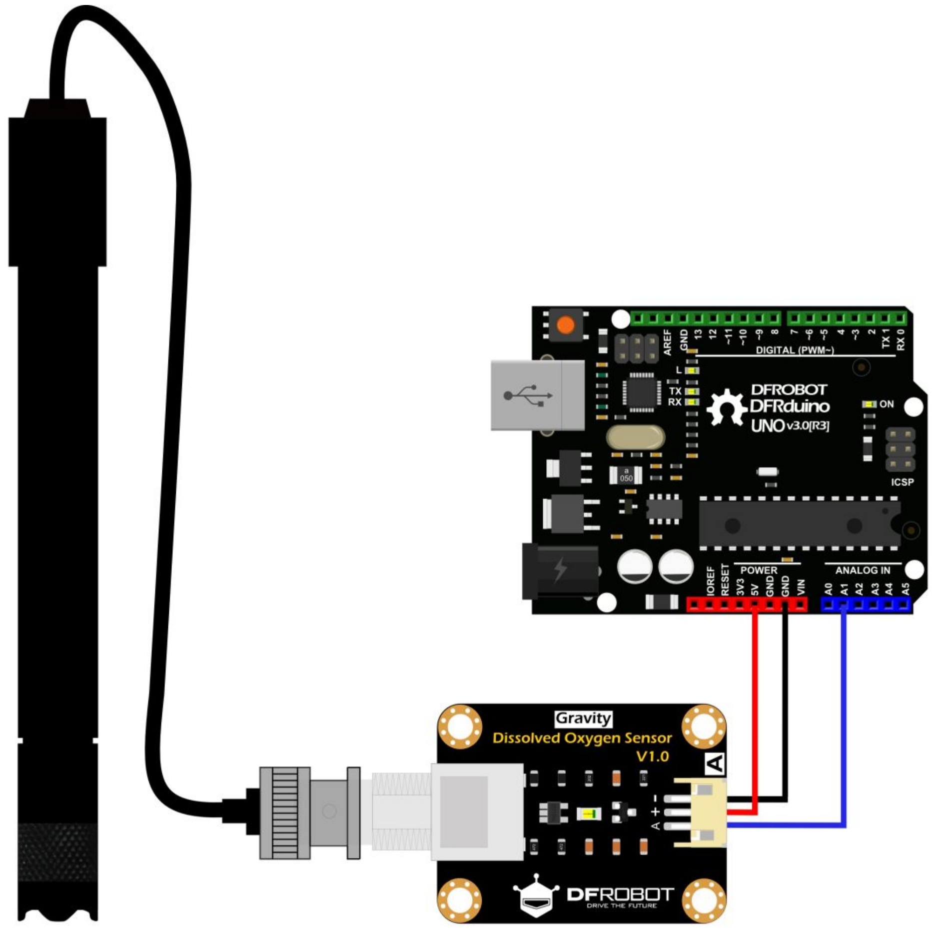
Steam Sensor

Soil Moisture Sensor



Ph and Alkalinity

Dissolved Oxygen Sensor (DO)



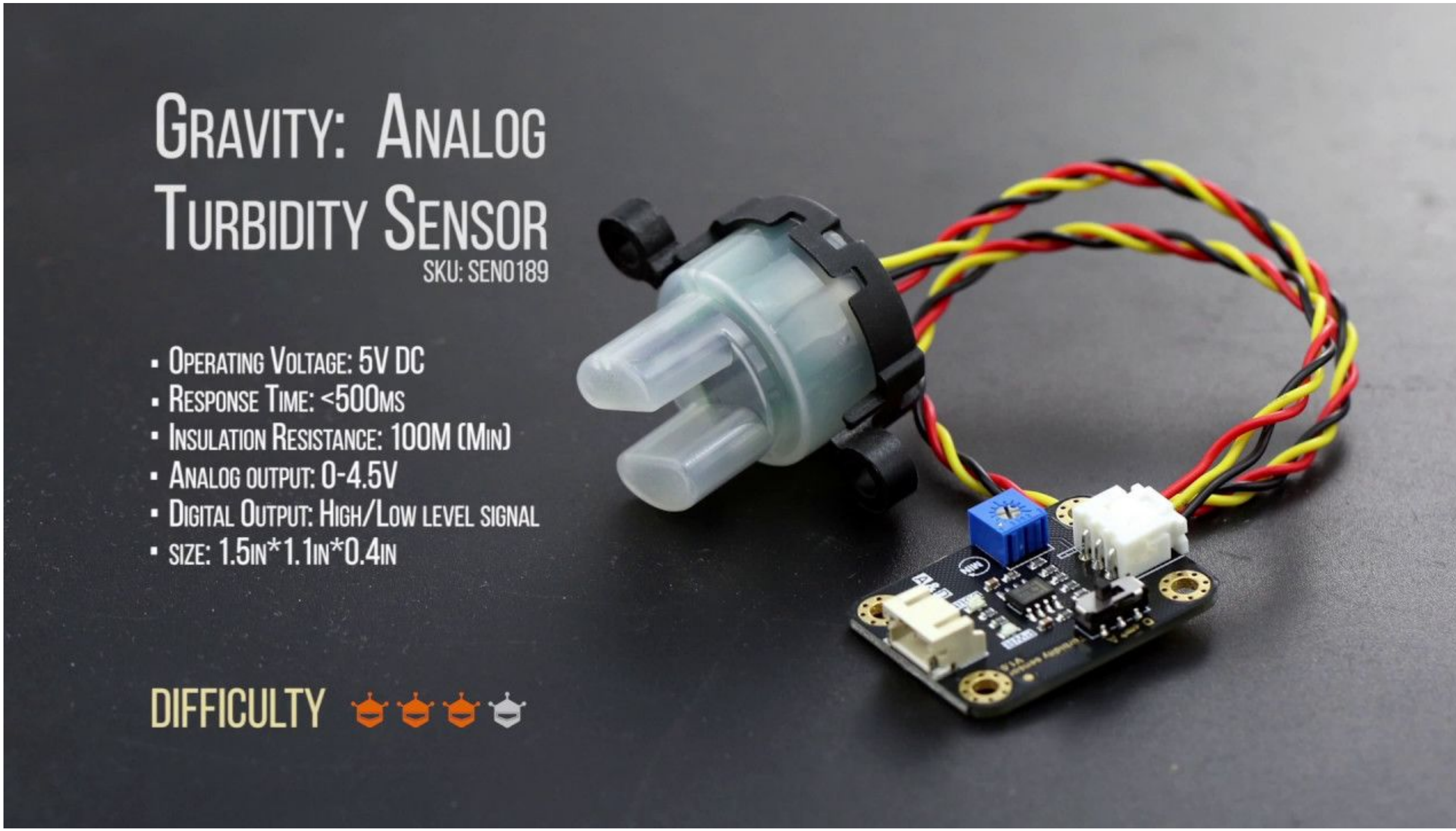
Biochemical Oxygen Demand (BOD)

BOD, COD, TSS

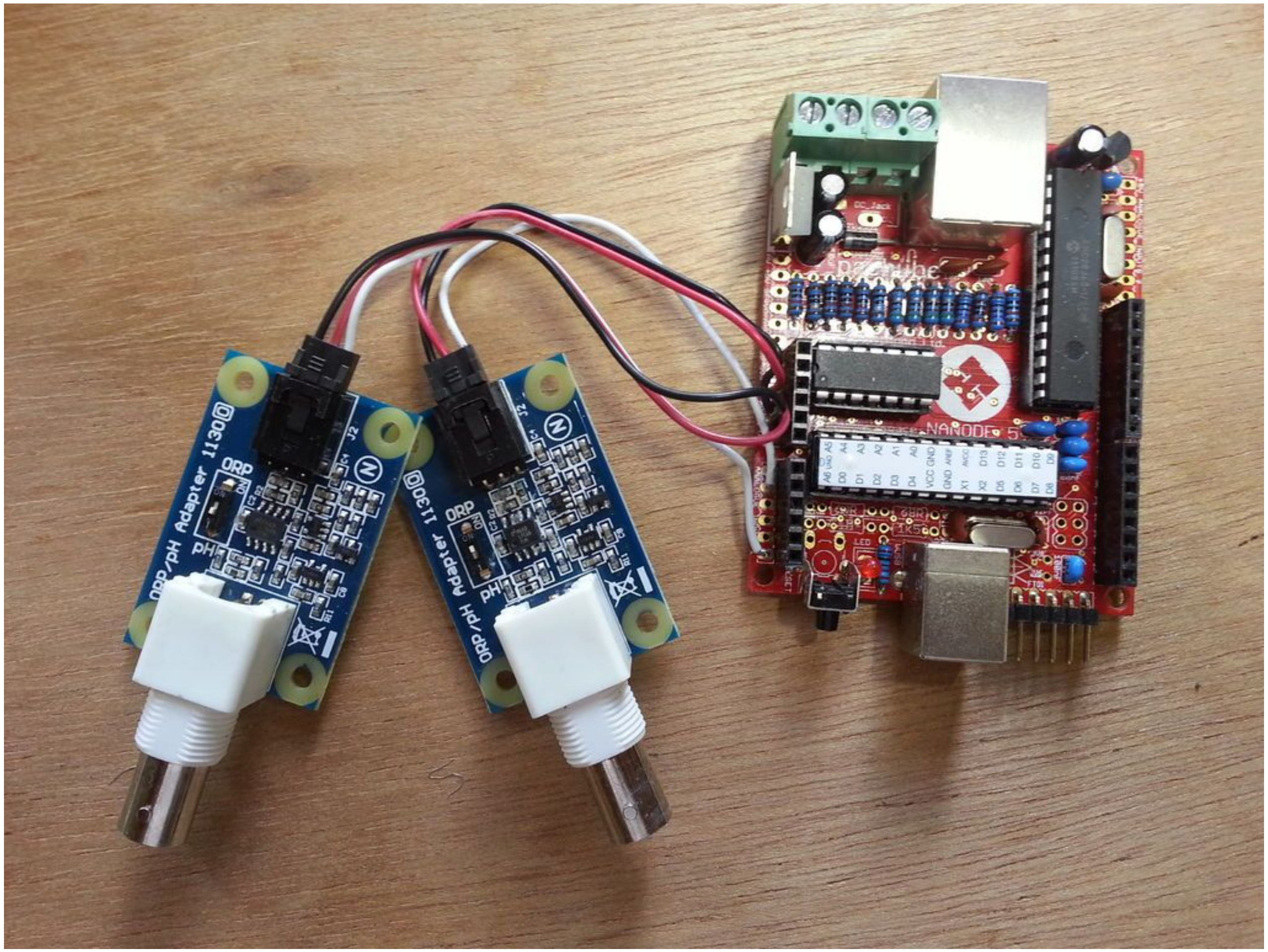
BOD: Biochemical Oxygen Demand
COD: Chemical Oxygen Demand
TSS: Turbidity And Total Suspended Solids

http://onlineeffluentmonitoring.com/bodcodtss/

Total Dissolved Solids S and Turbidity



Chlorine



Electrical Conductivity sensor water